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# ANCHORED SUPERVISED FINE-TUNING
# 锚定监督微调

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## ABSTRACT
## 摘要

Post-training of large language models involves a fundamental trade-off between supervised fine-tuning (SFT), which efficiently mimics demonstrations but tends to memorize, and reinforcement learning (RL), which achieves better generaliza-tion at higher computational cost. Dynamic Fine-Tuning (DFT) recently emerged as a promising middle ground, reweighting SFT objectives with token probabili-ties and achieving improvements in certain reasoning domains, though it exhibits instability in other tasks. We provide an analysis of DFT through the **reward-weighted regression (RWR)** framework, revealing that it corresponds to a spe-cific auxiliary distribution choice that yields provably tighter RL bounds than standard SFT. However, our analysis also uncovers a critical limitation: this con-struction lacks distributional anchoring, leading to progressive drift that under-mines training stability. To address this, we propose **Anchored Supervised Fine-Tuning (ASFT)**, which augments DFT's reweighting with **lightweight KL regu-larization** to preserve tightness while ensuring stability. Empirically, ASFT con-sistently outperforms both SFT and DFT across mathematical reasoning, medical knowledge grounding, and code generation, achieving substantial improvements with minimal computational overhead. Our RWR framework provides a system-atic lens for understanding post-training methods and demonstrates that principled theoretical analysis leads to both stronger guarantees and practical gains.[1]

大语言模型的微调涉及一个根本性的权衡：监督微调（SFT）虽然能高效地模仿示范，但容易记忆；而强化学习（RL）虽然计算成本更高，但能实现更好的泛化能力。最近，动态微调（DFT）作为一种有前景的中间方法出现，它通过使用 token 概率对 SFT 目标进行重新加权，在某些推理领域取得了改进，但其在其他任务中表现出不稳定性。我们通过奖励加权回归（RWR）框架对 DFT 进行了分析，发现它对应于一种特定的辅助分布选择，这种选择能够比标准 SFT 获得可证明更紧的 RL 界限。然而，我们的分析也揭示了一个关键的限制：这种构造缺乏分布锚定，导致训练稳定性受到破坏。为了解决这个问题，我们提出了锚定监督微调（ASFT），它在 DFT 的重新加权基础上添加了轻量级 KL 正则化，以保持紧致性并确保稳定性。实验证明，ASFT 在数学推理、医学知识基础和代码生成等任务中，相比 SFT 和 DFT 始终表现出色，取得了显著的改进，且计算开销极小。我们的 RWR 框架

为理解微调方法提供了一个系统的视角，并表明基于原理的理论分析能够带来更强的保证和实际收益。

## 1 INTRODUCTION

## 1 引言

Large language models (LLMs) have become a central substrate for modern AI systems, powering instruction following, tool use, and multi-step reasoning at scale (Brown et al., 2020; Touvron et al., 2023; Achiam et al., 2023; Guo et al., 2025). Post-training is crucial to adapt pretrained models to tasks and human preferences. This process typically involves two primary paradigms: supervised fine-tuning (SFT), which is an off-policy method that imitates expert demonstrations collected from a fixed dataset, and reinforcement learning (RL), which is an on-policy approach that optimizes outcome-based rewards by directly interacting with the model's own outputs (Ouyang et al., 2022; Rafailov et al., 2023; Shao et al., 2024). While SFT is data- and compute-efficient, excelling at rapid acquisition of desired behaviors, it tends to memorize surface patterns rather than learn robust, generalizable strategies (Chu et al., 2025; Zhang et al., 2021; Feldman, 2020). In contrast, RL leverages outcome-driven updates and exploration to discover more transferable behaviors, but is substantially more expensive and unstable in practice (Schulman et al., 2017; Ziegler et al., 2019; Stiennon et al., 2020). This fundamental trade-off motivates methods that retain SFT's efficiency while inheriting RL's generalization benefits (Bai et al., 2022; Lee et al., 2023; Yuan et al., 2024).

大语言模型（LLMs）已成为现代人工智能系统的核心基础，能够大规模地实现指令遵循、工具使用和多步骤推理（Brown 等，2020；Touvron 等，2023；Achiam 等，2023；Guo 等，2025）。后训练对于将预训练模型适应于任务和人类偏好至关重要。这一过程通常涉及两种主要范式：监督微调（SFT），这是一种离策略方法，通过模仿从固定数据集中收集的专家演示来实现；以及强化学习（RL），这是一种基于策略的方法，通过直接与模型自身的输出进行交互来优化基于结果的奖励（Ouyang 等，2022；Rafailov 等，2023；Shao 等，2024）。虽然 SFT 在数据和计算效率方面表现优异，能够快速获取期望的行为，但它往往只是记忆表面模式，而不是学习稳健且可泛化的策略（Chu 等，2025；Zhang 等，2021；Feldman，2020）。相比之下，RL 利用结果驱动的更新和探索来发现更具迁移性的行为，但在实践中成本更高且更不稳定（Schulman 等，2017；Ziegler 等，2019；Stiennon 等，2020）。这种根本性的权衡促使了那些在保持 SFT 高效性的同时继承 RL 泛化优势的方法（Bai 等，2022；Lee 等，2023；Yuan 等，2024）。

A growing body of work re-examines SFT through an RL lens, arguing that the *implicit reward* induced by maximum likelihood is pathological and that principled reweighting or trust regions are needed. Among these approaches, Dynamic Fine-Tuning (DFT) (Wu et al., 2025a) has gained significant attention by identifying a pathological reward structure in standard SFT that leads to unbounded variance when model probabilities approach zero. DFT addresses this through probability-based reweighting, achieving remarkable empirical improvements in mathematical reasoning tasks. However, our preliminary experiments reveal that DFT's effectiveness is domain-specific; it excels

越来越多的研究从强化学习（RL）的角度重新审视监督微调（SFT），认为最大似然诱导的隐式奖励是病态的，需要采用有原则的重新加权或信任区域方法。在这些方法中，动态微调（DFT）（Wu 等，2025a）通过识别标准 SFT 中的病态奖励结构而受到广泛关注，该结构会导致当模型概率接近零时方差无界。DFT 通过基于概率的重新加权来解决这一问题，在数学推理任务中取得了显著的实证改进。然而，我们的初步实验表明，DFT 的有效性具有领域特定性；它在某些领域表现尤为出色。

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[1]The code is available athttps://github.com/zhuchichi56/ASFT.

[1] 该代码可在以下链接获取 https://github.com/zhuchichi56/ASFT.

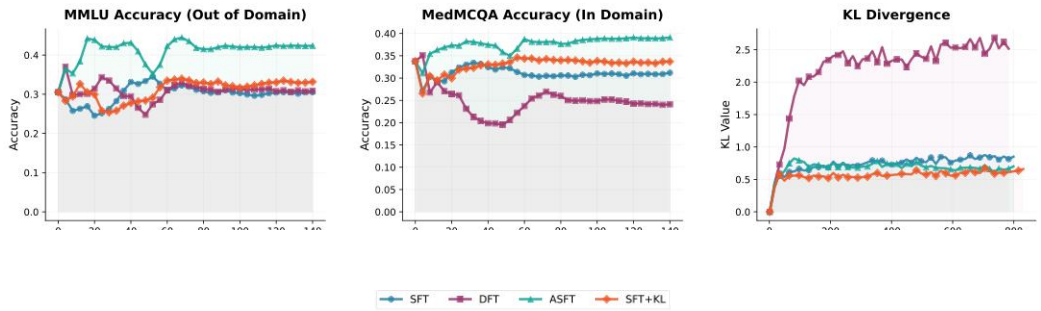


Figure 1: Training dynamics comparison across fine-tuning methods on medical knowledge tasks. **Left**: MMLU accuracy (out-of-domain evaluation); **Center**: MedMCQA accuracy (in-domain evaluation); **Right**: KL divergence from base model. DFT exhibits severe distributional drift (high KL divergence) while ASFT maintains stability through KL anchoring and achieves superior performance on both tasks.

图 1：在医学知识任务上不同微调方法的训练动态比较。左：MMLU 准确率（领域外评估）；中：MedMCQA 准确率（领域内评估）；右：与基线模型的 KL 散度。DFT 表现出严重的分布漂移（高 KL 散度），而 ASFT 通过 KL 锚定保持稳定，并在两个任务上均表现出色。

in reasoning-intensive domains yet exhibits instability in knowledge-intensive tasks and lacks theoretical grounding for its design choices.

在需要深入推理的领域中表现良好，但在知识密集型任务中表现出不稳定，并且其设计选择缺乏理论依据。

To address these problems, we provide a principled theoretical analysis of DFT within a reward-weighted regression framework inspired by reward-weighted regression and importance sampling theory (Rubinstein & Kroese, 2016). From the theoretical perspective, we reveal that DFT corresponds to a specific auxiliary distribution construction that yields a provably tighter lower bound on the RL objective compared to standard SFT. This causes a critical limitation: the absence of distributional anchoring mechanisms leads to progressive drift away from the reference distribution. The distributional shift makes the bound increasingly loose and increases importance weight variance, explaining the training instabilities of DFT.

为了解决这些问题，我们基于奖励加权回归和重要性采样理论（Rubinstein & Kroese, 2016）提供了一个以奖励加权回归框架为基础的理论分析。从理论角度来看，我们揭示了 DFT 对应于一种特定的辅助分布构造，相比标准的 SFT，它能够为强化学习目标提供可证明更紧的下界。这导致了一个关键的局限性：缺乏分布锚定机制会导致逐步偏离参考分布。分布偏移使得界限变得越来越松散，并增加重要性权重的方差，解释了 DFT 训练的不稳定性。

To address these fundamental limitations, we propose **Anchored Supervised Fine-Tuning (ASFT)**, a lightweight extension of DFT that incorporates a KL divergence regularization term to prevent distributional drift while preserving the tightness benefits of adaptive reweighting. As demonstrated in Figure 1, ASFT maintains stable KL divergence while achieving superior performance across both in-domain and out-of-domain evaluations.

为了解决这些基本限制，我们提出了锚定监督微调（ASFT），这是 DFT 的一个轻量级扩展，通过引入 KL 散度正则化项来防止分布漂移，同时保留自适应重加权的紧密性优势。如图 1 所示，ASFT 在保持稳定 KL 散度的同时，在领域内和领域外的评估中均实现了优越的性能。

Empirically, ASFT consistently outperforms both standard SFT and DFT across mathematical reasoning, medical knowledge, and code generation tasks. On mathematical reasoning benchmarks with 100k training samples, ASFT achieves an average improvement of +4.85 points (18.6%) over DFT and +17.89 points (142%) over the base model. In medical knowledge tasks with 10k samples, ASFT delivers +8.28 points (24.8%) improvement over SFT and +10.65 points (33.9%) over the base model, requiring only 3% of the training cost of full RL approaches.

从经验来看，ASFT 在数学推理、医学知识和代码生成任务中，始终优于标准的 SFT 和 DFT。在包含 10 万个训练样本的数学推理基准测试中，ASFT 比 DFT 平均提升 4.85 分（18.6%），比基础模型提升 17.89 分（142%）。在包含 1 万个样本的医学知识任务中，ASFT 相比 SFT 提升 8.28 分（24.8%），相比基础模型提升 10.65 分（33.9%），仅需全强化学习方法 3%的训练成本。

**Our contributions** are threefold: **(1)** We provide a rigorous theoretical explanation for DFT's domain-specific effectiveness and inherent instabilities, grounding its heuristic design within the for-mal reward-weighted regression framework and proving that it achieves a strictly tighter bound than SFT while suffering from uncontrolled variance growth. **(2)** We propose ASFT, a simple yet prin-cipled method that resolves DFT's stability issues through lightweight KL anchoring while main-taining its tightness advantages, requiring minimal computational overhead compared to full RL approaches. **(3)** We demonstrate that ASFT delivers superior performance across both reasoning-intensive and knowledge-intensive domains, achieving better generalization than SFT, greater sta-bility and broader applicability than DFT, and RL-comparable performance with SFT-level compu-tational efficiency.

我们的贡献主要有三个方面：(1) 我们提供了对 DFT 在特定领域有效性及其内在不稳定性进行严格理论解释，将其启发式设计建立在形式化的奖励加权回归框架内，并证明其在遭受不受控方差增长的情况下，所获得的界限比 SFT 更加严格。(2) 我们提出了 ASFT，这是一种简单而有理论依据的方法，通过轻量级 KL 锚定解决 DFT 的稳定性问题，同时保持其紧致性优势，与完整的强化学习方法相比，其计算开销极小。(3) 我们证明了 ASFT 在需要大量推理和知识的领域中均表现出色，其泛化能力优于 SFT，稳定性与适用性均优于 DFT，并且在计算效率与 SFT 相当的情况下，其性能可与强化学习相媲美。

## 2 Related Work

## 2 相关工作

**Supervised fine-tuning and reinforcement learning.** Supervised fine-tuning (SFT) and reinforcement learning (RL) are the two dominant paradigms for post-training large language models. SFT can be viewed as optimizing a stable but *loose lower bound* on the RL objective, which ensures efficiency and robustness but often leads to memorization and limited generalization (Wei et al.,

监督微调和强化学习。监督微调（SFT）和强化学习（RL）是训练大型语言模型后的主要范式。SFT 可以看作是对 RL 目标的一个稳定但较松的下界进行优化，这确保了效率和鲁棒性，但常常导致记忆化和泛化能力有限（Wei 等，

2022; Chung et al., 2024; Zhang et al., 2021; Chu et al., 2025). In contrast, RL directly optimizes outcome-based rewards, achieving tighter alignment and stronger generalization but at the cost of instability, high variance, and heavy computation (Ouyang et al., 2022; Schulman et al., 2015; 2017). Recent work seeks to bridge this trade-off by either tightening the SFT bound via reweighting and importance weighting (Wu et al., 2025a; Qin & Springenberg, 2025) or stabilizing RL through trust-region and hybrid methods (Sheng et al., 2025; Zhu et al., 2025).

2022 年；Chung 等，2024 年；Zhang 等，2021 年；Chu 等，2025 年）。相比之下，强化学习（RL）直接优化基于结果的奖励，从而实现更紧密的对齐和更强的泛化能力，但代价是不稳定、高方差和计算量大（Ouyang 等，2022 年；Schulman 等，2015 年；2017 年）。最近的研究试图通过重新加权和重要性加权来收紧 SFT 边界（Wu 等，2025a；Qin & Springenberg, 2025），或者通过信任区域和混合方法来稳定强化学习（Sheng 等，2025；Zhu 等，2025）。

**Importance weighting and policy optimization.** The connection between supervised learning and reinforcement learning through importance weighting has deep theoretical roots (Kahn & Mar-shall, 1953; Dayan & Hinton, 1997). In the context of policy optimization, importance sampling enables off-policy learning by reweighting samples from a behavior policy to estimate gradients for a target policy (Metelli et al., 2018; Jiang & Li, 2016), though the resulting weights can suf-fer from high variance when distributions are misaligned (Andrad´ottir et al., 1995). To address this, trust-region methods (Schulman et al., 2015) and proximal policy optimization (Schulman et al., 2017) constrain policy updates to remain close to a reference policy. Building on these ideas, recent advances in language model fine-tuning have explored importance-weighted super-vised objectives (Qin & Springenberg, 2025), proximal supervised fine-tuning (Zhu et al., 2025), and probability-based reweighting in dynamic fine-tuning (Wu et al., 2025a), as well as unified frameworks that combine SFT and RL principles (Lv et al., 2025; Wu et al., 2025b). Our work follows this line of research by anchoring the auxiliary distribution itself to the base model, thereby extending importance-weighted approaches with a mechanism that stabilizes training while retain-ing their theoretical tightness.

重要性加权和策略优化。通过重要性加权将监督学习与强化学习联系起来具有深厚的理论基础（Kahn & Marshall, 1953; Dayan & Hinton, 1997）。在策略优化的背景下，重要性抽样通过重新加权行为策略的样本以估计目标策略的梯度，从而实现离策略学习（Metelli 等, 2018; Jiang & Li, 2016），尽管当分布不匹配时，所产生的权重可能会出现高方差（Andrad ó ttir 等, 1995）。为了解决这个问题，信任区域方法（Schulman 等, 2015）和邻近策略优化（Schulman 等, 2017）限制策略更新以保持与参考策略的接近。基于这些思想，最近在语言模型微调方面的进展探索了重要性加权的监督目标（Qin & Springenberg, 2025）、邻近监督微调（Zhu 等, 2025）以及动态微调中的基于概率的重加权（Wu 等, 2025a），以及结合 SFT 和 RL 原理的统一框架（Lv 等, 2025; Wu 等, 2025b）。我们的工作沿着这一研究路线展开，将辅助分布本身锚定到基础模型上，从而通过一种稳定训练同时保持其理论紧致性的机制扩展了重要性加权方法。

## 3 PRELIMINARIES

## 3 预备知识

In this section, we establish the theoretical foundation connecting supervised fine-tuning and reinforcement learning through the reward-weighted regression framework, setting the stage for our analysis of existing methods and our proposed ASFT approach.

在本节中，我们通过奖励加权回归框架，建立了监督微调与强化学习之间的理论基础，为后续对现有方法的分析以及我们提出的 ASFT 方法奠定了基础。

### 3.1 Problem Formulation and Basic Frameworks

3.1 问题建模与基本框架

We consider language modeling where trajectories $\tau = (x, y)$ consist of input prompts $x$ and generated responses $y$. A parametric policy $\pi_\theta(\tau) = \pi_\theta(y \mid x) = \prod_{|y|,\, t=1} \pi_\theta(y_t \mid y_{<t}, x)$ assigns probabil-ity via autoregressive decomposition. The **reinforcement learning objective** maximizes expected reward $J(\theta) = \mathrm{E}_{\tau \sim \pi_\theta}[R(\tau)]$ where $R(\tau) : X \times Y \to [0, 1]$ evaluates trajectory quality. In contrast, **supervised fine-tuning** performs behavior cloning on expert demonstrations $D = \{(x, y_*)\}$ sam-pled from reference distribution $\pi_{\mathrm{ref}}(\tau)$ by minimizing $L_{\mathrm{SFT}}(\theta) = -\mathrm{E}_{(x,y_*) \sim D}[\log \pi_\theta(y_* \mid x)]$.

我们考虑一种语言建模方法，其中轨迹 $\tau = (x, y)$ 由输入提示 x 和生成的响应 y 组成。一个参数化策略 $\pi_\theta(\tau) = \pi_\theta(y \mid x) = \prod_{|y|,\, t=1} \pi_\theta(y_t \mid y_{<t}, x)$ 通过自回归分解来分配概率。强化学习的目标是最大化预期奖励 $J(\theta) = \mathrm{E}_{\tau \sim \pi_\theta}[R(\tau)]$，其中 $R(\tau) : X \times Y \to [0, 1]$ 评估轨迹的质量。相比之下，监督微调通过最小化 $L_{\mathrm{SFT}}(\theta) = -\mathrm{E}_{(x,y_*) \sim D}[\log \pi_\theta(y_* \mid x)]$ 来对专家演示 $D = \{(x, y_*)\}$ 进行行为克隆，这些演示是从参考分布 $\pi_{\mathrm{ref}}(\tau)$ 中采样的。

### 3.2 The Reward-Weighted Regression Framework

3.2 奖励加权回归框架

Building on prior work in reward-weighted regression and importance sampling (Peters & Schaal, 2007; Rubinstein & Kroese, 2016; Qin & Springenberg, 2025), we adopt the **reward-weighted regression (RWR) framework** for language model fine-tuning. This framework provides a principled connection between SFT and RL objectives by leveraging importance sampling and auxiliary distributions to construct tighter bounds on the RL objective.

在奖励加权回归和重要性采样（Peters & Schaal, 2007; Rubinstein & Kroese, 2016; Qin & Springenberg, 2025）方面的先前工作基础上，我们采用奖励加权回归（RWR）框架来进行语言模型的微调。该框架通过利用重要性采样和辅助分布，建立了一个有理论依据的连接，将监督微调（SFT）目标与强化学习（RL）目标联系起来，并构建对 RL 目标更紧的界限。

Under sparse rewards where $R(\tau) = \mathrm{I}[y = y_*]$ and the assumption that $\mathrm{supp}(\pi_\theta) \subseteq \mathrm{supp}(\pi_{\mathrm{ref}})$, we can establish the following fundamental result:

在稀疏奖励情况下，其中 $R(\tau) = \mathrm{I}[y = y_*]$，并且假设 $\mathrm{supp}(\pi_\theta) \subseteq \mathrm{supp}(\pi_{\mathrm{ref}})$，我们可以得出以下基本结论：

**Proposition 1** (SFT as RL Lower Bound)**.** *The RL objective satisfies:*

命题 1（SFT 作为 RL 下界）。RL 目标满足：

$$J(\theta) \geq c_{\mathrm{ref}} \cdot \mathrm{E}_{\tau \in D_+}[\log \pi_\theta(\tau)] \quad (1)$$

*where* $D_+ = \{(x, y_*) \mid R(x, y_*) = 1\}$ *and* $c_{\mathrm{ref}} = \mathrm{P}_{\pi_{\mathrm{ref}}}(\tau \in D_+)$.

其中 $D_+ = \{(x, y_*) \mid R(x, y_*) = 1\}$，$c_{\mathrm{ref}} = \mathrm{P}_{\pi_{\mathrm{ref}}}(\tau \in D_+)$。

This reveals that SFT optimization implicitly maximizes a lower bound on the RL objective. However, this bound becomes increasingly loose as $\pi_\theta$ diverges from $\pi_{\text{ref}}$ during training.

这表明 SFT 优化实际上是在隐式地最大化对 RL 目标的一个下界。然而，随着训练过程中 $\pi_\theta$ 偏离 $\pi_{\text{ref}}$，这个界限会变得越来越松散。

Within the RWR framework, we can generalize to tighter bounds through auxiliary distributions.

在 RWR 框架中，我们可以通过辅助分布来推广得到更紧的界限。

For any distribution $q(\tau)$ with appropriate support:

$$J(\theta) \geq c_{\text{ref}} \cdot \mathrm{E}_{\tau \in D_+} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} \log \pi_\theta(\tau) \tag{2}$$

对于任何具有适当支撑集的分布 q(τ)：

$$J(\theta) \geq c_{\text{ref}} \cdot \mathrm{E}_{\tau \in D_+} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} \log \pi_\theta(\tau) \tag{2}$$

The choice of auxiliary distribution $q$ determines both the tightness of the bound and the stability of the resulting optimization procedure. This sets up the fundamental trade-off between validity and tightness that we address in this work.

辅助分布 q 的选择决定了界限的紧密程度以及由此产生的优化过程的稳定性。这确立了我们在本工作中所研究的有效性与紧密程度之间的基本权衡。

### 3.3 Dynamic Fine-Tuning: An Existing Approach

### 3.3 动态微调：一种现有方法

Dynamic Fine-Tuning (DFT) (Wu et al., 2025a) addresses SFT's limitations by identifying a pathological reward structure in standard SFT. When viewed as a policy gradient method, SFT's implicit reward $r_{\text{SFT}}(y \mid x) = \mathbb{I}[y=y_*]/\pi_\theta(y|x)$ exhibits inverse-probability weighting that causes unbounded variance when $\pi_\theta(y_* \mid x)$ approaches zero.

动态微调（DFT）（Wu 等，2025a）通过识别标准 SFT 中的病理奖励结构，解决了 SFT 的局限性。当将 SFT 视为策略梯度方法时，其隐式奖励 $r_{\text{SFT}}(y \mid x) = \mathbb{I}[y=y_*]/\pi_\theta(y|x)$ 表现出逆概率加权，当 $\pi_\theta(y_* \mid x)$ 接近零时会导致方差无界。

DFT addresses this through probability-based reweighting:

DFT 通过基于概率的重新加权来解决这个问题：

$$L_{\text{DFT}}(\theta) = -\mathrm{E}_{(x,y_*)\sim D}[\text{sg}[\pi_\theta(y_* \mid x)] \log \pi_\theta(y_* \mid x)] \tag{3}$$

$$L_{\text{DFT}}(\theta) = -\mathrm{E}_{(x,y_*)\sim D}[\text{sg}[\pi_\theta(y_* \mid x)] \log \pi_\theta(y_* \mid x)] \tag{3}$$

where sg[·] denotes the stop-gradient operator.

其中 sg[·] 表示停止梯度操作符。

While DFT achieves empirical improvements, its theoretical properties within the RWR framework remained unclear—a gap we address in our analysis.

尽管 DFT 在实践中取得了改进，但在 RWR 框架下的理论性质仍不清楚——这是我们分析中要解决的不足之处。

## 4 Method

## 4 方法

### 4.1 Theoretical Analysis of DFT within the RWR Framework

4.1 在 RWR 框架内的 DFT 理论分析

We found that DFT can be precisely characterized within the RWR framework through a specific auxiliary distribution construction. This analysis reveals both DFT's strengths and fundamental limitations.

我们发现，通过构建特定的辅助分布，可以在 RWR 框架内精确地描述 DFT。这一分析揭示了 DFT 的优势及其基本局限性。

**Key Finding 1: DFT corresponds to a specific auxiliary distribution choice.** We discovered that the DFT objective is mathematically equivalent to choosing the auxiliary distribution:

关键发现 1：DFT 对应于特定的辅助分布选择。我们发现 DFT 的目标在数学上等同于选择辅助分布：

$$q(\tau) = \pi_{\text{ref}}(\tau \mid D_+)\ \text{sg}[p_\theta(\tau)]\ (4)\ \mathrm{E}_{\tau \sim \pi_{\text{ref}}(\cdot \mid D_+)}[\text{sg}[p_\theta(\tau)]]$$

q(τ) = πref(τ | D+) sg[pθ(τ)] (4) Eτ∼πref(· |D+)[sg[pθ(τ)]]

This construction directly recovers the DFT sequence-level objective:

这种构造直接恢复了 DFT 序列级别的目标：

$$L_{\text{DFT}}(\theta) = -\mathrm{E}_{\tau \in D_+}[\text{sg}(p_\theta(\tau)) \log p_\theta(\tau)] \tag{5}$$

**Key Finding 2: DFT achieves provably tighter bounds than SFT.** We proved that this auxil-iary distribution yields a strictly tighter lower bound on the RL objective compared to standard SFT whenever the policy assigns non-uniform probabilities to demonstrations (detailed proof in Appendix D.4).

关键发现 2：DFT 在证明上比 SFT 获得更紧的界限。我们证明了，当策略对演示分配非均匀概率时，这种辅助分布会比标准的 SFT 产生更严格的下界（详细证明见附录 D.4）。

**Theorem 1** (Strict Tightness)**.** *The DFT auxiliary distribution yields a strictly tighter lower bound than standard SFT whenever* $\mathrm{Var}(p_\theta(\tau)) > 0$ *on* $D_+$.

定理 1（严格紧致性）。当在 D+上 Var(pθ(τ)) > 0 时，DFT 辅助分布产生的下界比标准 SFT 更严格。

This theoretical result explains DFT's superior empirical performance in domains where the policy distribution exhibits sufficient variance across training examples.

这一理论结果解释了 DFT 在策略分布在整个训练样本中表现出足够方差的领域中，具有优越的实证性能。

**Key Finding 3: DFT suffers from distributional drift.** However, our analysis also revealed a critical limitation: the policy distribution progressively diverges from the reference distribution during training. As optimization proceeds, $q$ becomes increasingly concentrated on trajectories with high $p_\theta(\tau)$, creating a feedback loop where the model focuses on a diminishing subset of training data. This distributional drift makes the bound increasingly loose and the importance weights increasingly high-variance, reducing effective sample size and destabilizing optimization.

关键发现 3：DFT 存在分布漂移的问题。然而，我们的分析也揭示了一个关键的限制：在训练过程中，策略分布逐渐偏离参考分布。随着优化的进行，q 越来越集中在具有高 pθ(τ)的轨迹上，形成一个反馈循环，使得模型越来越关注训练数据中越来越少的一小部分。这种分布漂移使得界变得越来越松散，重要权重的方差也越来越大，从而降低了有效样本量并使优化变得不稳定。

We formalized this instability by noting that the fundamental inequality $u \geq 1 + \log u$ (used to derive the RL lower bound) achieves equality if and only if $u = 1$. In DFT's case, $u = \pi_\theta(\tau)/q_\theta(\tau)$, so the bound is tight only when $\pi_\theta(\tau) = q_\theta(\tau)$, i.e., when $p_\theta(\tau)$ is constant on $D_+$. However, as training progresses, $p_\theta(\tau)$ becomes increasingly non-uniform, making the inequality strictly loose.

我们通过指出基本不等式 $u \geq 1 + \log u$（用于推导 RL 下界）仅当 $u = 1$ 时取等号，从而形式化了这种不稳定性。在 DFT 的情况下，$u = \pi_\theta(\tau)/q_\theta(\tau)$，因此只有当 $\pi_\theta(\tau) = q_\theta(\tau)$，即 $p_\theta(\tau)$ 在 $D_+$ 上为常数时，该界才紧致。然而，随着训练的进行，$p_\theta(\tau)$ 越来越不均匀，使得该不等式变得严格宽松。

This leads to deteriorating bound quality, reduced effective sample size, and training instability.

这会导致界限质量下降、有效样本量减少以及训练不稳定。

### 4.2 Anchored Supervised Fine-Tuning (ASFT)

### 4.2 锚定监督微调（ASFT）

To address DFT's distributional drift while preserving its tightness advantages, we propose **Anchored Supervised Fine-Tuning (ASFT)**. Our method adds a lightweight KL regularization term that constrains the policy within a trust region of a reference checkpoint:

为了解决 DFT 的分布漂移问题，同时保持其紧致性优势，我们提出了一种名为 Anchored Supervised Fine-Tuning（ASFT）的方法。我们的方法添加了一个轻量级的 KL 正则化项，该正则化项将策略限制在参考检查点的信任区域内：

$$L_{\mathrm{ASFT}}(\theta) = L_{\mathrm{DFT}}(\theta) + \lambda \mathbb{E}_s[D_{\mathrm{KL}}(\pi_{\mathrm{base}}(\cdot \mid s) \,/\!/\, \pi_\theta(\cdot \mid s))] \tag{6}$$

where $\pi_{\mathrm{base}}$ is a fixed reference policy (typically the pretrained model) and $\lambda > 0$ controls anchoring strength.

其中 $\pi_{\mathrm{base}}$ 是一个固定的参考策略（通常为预训练模型），$\lambda > 0$ 用于控制锚定强度。

**Theoretical Guarantees.** This design preserves DFT's tightness benefits since the KL term does not alter the lower-bound structure, while providing explicit variance control that prevents the expo-nential growth that destabilizes pure DFT training. The anchoring mechanism creates a trust region around the reference policy, allowing controlled exploration of tighter bounds without sacrificing distributional stability.

理论保证。此设计保留了 DFT 的紧致性优势，因为 KL 项不会改变下界结构，同时提供了显式的方差控制，防止了导致纯 DFT 训练不稳定的指数增长。锚定机制在参考策略周围创建了一个信任区域，允许在不牺牲分布稳定性的情况下对更紧的界限进行受控探索。

**Practical Implementation.** Following standard practice in language model training (Ouyang et al., 2022; Shao et al., 2024), we implement ASFT at the token level by distributing sequence-level weights across tokens using normalized per-position allocation, ensuring mathematical equivalence to our theoretical framework while enabling efficient computation. Our method requires minimal computational overhead compared to standard SFT - adding only a simple KL penalty - yet delivers RL-comparable generalization performance along with SFT-level efficiency.

实用实现。遵循语言模型训练的标准实践（Ouyang 等人，2022；Shao 等人，2024），我们在标记级别实现 ASFT，通过使用归一化的每个位置分配将序列级权重分布到各个标记上，确保与我们的理论框架在数学上等价，同时实现高效计算。与标准 SFT 相比，我们的方法计算开销极

小——仅增加了一个简单的 KL 惩罚项，却能够实现与 RL 相当的泛化性能，同时保持 SFT 级别的效率。

# 5 EXPERIMENTS

# 5 实验

## 5.1 SETUP

## 5.1 设置

**Models.** We conduct fine-tuning experiments using LLaMA-2-7B(Touvron et al., 2023) and Qwen2.5-7B(Qwen et al., 2025), two widely adopted models in the field. We select LLaMA-2-7B specifically to avoid potential contamination from prior supervised knowledge. Qwen2.5-7B, on the other hand, is a state-of-the-art model that is broadly used in current research. For knowledge-intensive (medical) tasks, we utilize both LLaMA-2-7B and Qwen2.5-7B to evaluate knowledge fine-tuning. For mathematical reasoning tasks, we focus exclusively on Qwen2.5-7B due to its superior reasoning capabilities, whereas LLaMA-2-7B is less competitive in mathematical reasoning. This setup enables a systematic study of knowledge and reasoning learning across both fact-based and reasoning-intensive domains.

模型。我们使用 LLaMA-2-7B（Touvron 等，2023）和 Qwen2.5-7B（Qwen 等，2025）两种在该领域广泛应用的模型进行微调实验。我们特别选择 LLaMA-2-7B 是为了避免潜在的先前监督知识污染。另一方面，Qwen2.5-7B 是一个最先进的模型，目前被广泛用于各种研究。对于知识密集型（如医疗）任务，我们使用 LLaMA-2-7B 和 Qwen2.5-7B 来评估知识微调的效果。而对于数学推理任务，我们则专注于 Qwen2.5-7B，因为其推理能力更优越，而 LLaMA-2-7B 在数学推理方面竞争力较弱。这种设置使我们能够系统地研究基于事实和推理密集型领域的知识与推理学习。

**Datasets.** We evaluate ASFT on two domains: (i) **Mathematical reasoning**, using 10k/30k/100k samples from NuminaMath CoT (LI et al., 2024) for training, and testing on Math500 (Hendrycks et al., 2021), Minerva Math (Lewkowycz et al., 2022), OlympiadBench (AI Mathematical Olympiad, 2024), AIME 2024 (American Institute of Mathematics, 2024), and AMC 2023 (Mathematical As-sociation of America, 2023); (ii) **Medical knowledge**, using 10k/30k/100k MedMCQA (Pal et al., 2022) samples for training, and testing on MMLU-medical (Hendrycks et al., 2020), MedQA (Jin et al., 2021), and the MedMCQA test set.

数据集。我们在两个领域上评估 ASFT：(i) 数学推理，使用来自 NuminaMath CoT (LI 等，2024) 的 10k/30k/100k 样本进行训练，并在 Math500 (Hendrycks 等，2021)、Minerva Math (Lewkowycz 等，2022)、OlympiadBench (AI Mathematical Olympiad, 2024)、AIME 2024 (American Institute of Mathematics, 2024)和 AMC 2023 (Mathematical Association of America, 2023)上进行测试；(ii) 医疗知识，使用来自 MedMCQA (Pal 等，2022)的 10k/30k/100k 样本进行训练，并在 MMLU-medical (Hendrycks 等，2020)、MedQA (Jin 等，2021)以及 MedMCQA 测试集上进行测试。

**Training and Evaluation Settings.** All methods are implemented using AdamW optimizer, co-sine learning rate decay, and warm-up ratio 0.1. For **mathematical reasoning**, SFT and DFT use modelmaxlength 2048, globalbatchsize 256, learningrate $5 \times 10^{-5}$, and are trained for 1 epoch. ASFT follows the same configuration with coefficient $\lambda = 0.05$. For **medical knowledge**, we set modelmaxlength 512, globalbatchsize 64, learningrate $2 \times 10^{-5}$, and train for 3 epochs, with ASFT again using $\lambda = 0.05$. At the evaluation stage, for math, we use the default chat template

训练和评估设置。所有方法均使用 AdamW 优化器、余弦学习率衰减和 warm-up 比例 0.1。对于数学推理，SFT 和 DFT 使用模型最大长度 2048，全局批次大小 256，学习率 $5 \times 10^{-5}$ ，并训练 1

个周期。ASFT 遵循相同的配置，系数 $\lambda$ =0.05。对于医学知识，我们设置模型最大长度 512，全局批次大小 64，学习率 $2\times10^{-5}$ ，并训练 3 个周期，ASFT 同样使用 $\lambda$ =0.05。在评估阶段，对于数学问题，我们使用默认的聊天模板

| Medical Benchmarks<br>医学基准 | | | | | Math Benchmarks<br>数学基准 | | | | | | Methods<br>方法 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | MedQA | MMLU | MedMCQA | Avg. | AIME24 | Math500 | Minerva | Olympiad | ACM23 | Avg. | |
| Base | 29.85 | 30.52 | 33.76 | 31.38 | 1.65 | 28.79 | 9.26 | 7.69 | 15.65 | 12.61 | |
| ***Dataset Scale: 10k***<br>数据集规模：10k | | | | | | | | | | | |
| SFT | 33.31 ↑3.46 | 33.52 ↑3.00 | 33.28 ↓0.48 | 33.37 ↑1.99 | 1.24 ↓0.41 | 41.84 ↑13.05 | 11.30 ↑2.04 | 12.26 ↑4.57 | 17.03 ↑1.38 | 16.73 ↑4.12 | |
| SFT w/ KL | 29.22 ↓0.63 | 30.63 ↑0.11 | 33.01 ↓0.75 | 30.95 ↓0.43 | 0.41 ↓1.24 | 42.21 ↑13.42 | 12.05 ↑2.79 | 12.08 ↑4.39 | 17.19 ↑1.54 | 16.79 ↑4.18 | |
| DFT | 29.69 ↓0.16 | 26.69 ↓3.83 | 31.20 ↓2.56 | 29.19 ↓2.19 | **4.18** ↑2.53 | **59.51** ↑30.72 | 17.10 ↑7.84 | **24.95** ↑17.26 | 31.09 ↑15.44 | 27.77 ↑15.16 | |
| **ASFT** | **39.28** ↑9.43 | **46.37** ↑15.85 | **40.45** ↑6.69 | **42.03** ↑10.65 | 3.33 ↑1.68 | 59.60 ↑30.81 | **19.91** ↑10.65 | 24.50 ↑16.81 | **36.41** ↑20.76 | **28.75** ↑16.14 | |
| ***Dataset Scale: 30k***<br>数据集规模：30k | | | | | | | | | | | |
| SFT | 33.54 ↑3.69 | 38.48 ↑7.96 | 36.03 ↑2.27 | 36.02 ↑4.64 | 2.71 ↑1.06 | 44.74 ↑15.95 | 13.21 ↑3.95 | 13.44 ↑5.75 | 21.56 ↑5.91 | 19.93 ↑7.32 | |
| SFT w/ KL | 30.56 ↑0.71 | 29.86 ↓0.66 | 33.56 ↓0.20 | 31.33 ↓0.05 | 2.70 ↑1.05 | 44.91 ↑16.12 | 13.03 ↑3.77 | 13.48 ↑5.79 | 18.90 ↑3.25 | 18.60 ↑5.99 | |
| DFT | 31.26 ↑1.41 | 33.08 ↑2.56 | 35.09 ↑1.33 | 33.14 ↑1.76 | 3.34 ↑1.69 | **57.93** ↑29.14 | **23.28** ↑14.02 | **25.31** ↑17.62 | 28.44 ↑12.79 | **27.66** ↑15.05 | |
| **ASFT** | **42.03** ↑12.18 | **44.94** ↑14.42 | **39.06** ↑5.30 | **42.01** ↑10.63 | **5.81** ↑4.16 | 57.03 ↑28.24 | 20.61 ↑11.35 | 24.44 ↑16.75 | **30.00** ↑14.35 | 27.18 ↑14.57 | |
| ***Dataset Scale: 100k***<br>数据集规模：10 万 | | | | | | | | | | | |
| SFT | 33.46 ↑3.61 | 38.01 ↑7.49 | 35.67 ↑1.91 | 35.71 ↑4.33 | 0.83 ↓0.82 | 47.30 ↑18.51 | 13.46 ↑4.20 | 14.16 ↑6.47 | 20.00 ↑4.35 | 19.15 ↑6.54 | |
| SFT w/ KL | 30.09 ↑0.24 | 31.62 ↑1.10 | 33.85 ↑0.09 | 31.85 ↑0.47 | 1.44 ↓0.21 | 46.81 ↑17.02 | 14.13 ↑4.87 | 13.74 ↑6.05 | 20.00 ↑4.35 | 19.22 ↑6.61 | |
| DFT | 36.61 ↑6.76 | 41.26 ↑10.74 | 36.31 ↑2.55 | 38.06 ↑6.68 | 6.26 ↑4.61 | 56.88 ↑28.09 | 21.18 ↑11.92 | 22.68 ↑14.99 | 27.19 ↑11.54 | 26.04 ↑13.43 | |
| **ASFT** | **40.61** ↑10.76 | **42.02** ↑11.50 | **37.32** ↑3.56 | **39.98** ↑8.60 | **6.66** ↑5.01 | **59.99** ↑31.20 | **23.55** ↑14.29 | **25.57** ↑17.88 | **36.72** ↑21.07 | **30.50** ↑17.89 | |

Table 1: Performance comparison of fine-tuning methods on medical benchmarks (left, base: LLaMA-2-7B) and math benchmarks (right, base: Qwen2.5-7B) under different dataset scales. **Bold** numbers indicate the best performance in each group, and rows with blue background highlight our ASFT approach. Arrows with ↑and ↓indicate performance improvements and degradations relative to the base model.

表 1：在不同数据集规模下，微调方法在医学基准（左侧，基础模型：LLaMA-2-7B）和数学基准（右侧，基础模型：Qwen2.5-7B）上的性能比较。粗体数字表示每组中的最佳性能，背景为蓝色的行突出显示我们的 ASFT 方法。↑和↓箭头表示相对于基础模型的性能提升和下降。

and Chain-of-Thought (CoT) prompting, report average accuracy over 16 decoding runs (temperature 1.0, max length 4096). For medical, we use standard prompt templates and multiple-choice accuracy. Baselines include SFT, SFT w/ KL, and DFT (Wu et al., 2025b).

此外，使用思维链（Chain-of-Thought，CoT）提示，报告在 16 次解码运行中的平均准确率（温度 1.0，最大长度 4096）。对于医疗领域，我们使用标准提示模板和多项选择准确率。基线包括 SFT、带 KL 的 SFT 以及 DFT（Wu 等，2025b）。

### 5.2 Main Results

### 5.2 主要结果

Our experimental results demonstrate that ASFT consistently delivers superior performance across both knowledge-intensive and reasoning-intensive domains while maintaining training stability. As shown in Table 1, across both medical knowledge and mathematical reasoning tasks, ASFT consistently delivers strong and stable improvements over all baselines. In knowledge-intensive domains, ASFT not only avoids the severe performance degradation observed with DFT (which drops by an average of -2.19 points at 10k samples), but also achieves substantial gains—outperforming the

base model by +10.65 points (a 33.9% relative improvement) at 10k scale, and maintaining robust advan-tages as the dataset size increases (10k: +10.65, 30k: +10.63, 100k: +8.60). This stability across scales highlights ASFT 's scalability and addresses the distributional drift issues that limit DFT in such settings. For mathematical reasoning, both DFT and ASFT surpass standard SFT, but ASFT maintains a consistent edge and greater training stability. With 100k samples, ASFT improves over the base by +17.89 points (versus DFT's +13.43), and the advantage is even more pronounced on challenging benchmarks like AMC23 (36.72% vs. 27.19% for DFT), reflecting superior generaliza-tion. Overall, ASFT 's improvements are not only larger but also more consistent across diverse benchmarks, while DFT's gains are more variable—especially on tasks requiring broad mathemati-cal reasoning—underscoring the robustness and effectiveness of ASFT.

我们的实验结果表明，ASFT 在知识密集型和推理密集型领域中始终表现出色，同时保持了训练的稳定性。如表 1 所示，在医学知识和数学推理任务中，ASFT 在所有基线模型上都持续表现出强劲且稳定的提升。在知识密集型领域中，ASFT 不仅避免了 DFT 所观察到的严重性能下降（在 10,000 个样本时平均下降-2.19 分），还取得了显著的提升——在 10,000 个样本规模时，ASFT 优于基线模型+10.65 分（相对提升 33.9%），并且随着数据集规模的增加，仍能保持优势（10,000 个样本：+10.65，30,000 个样本：+10.63，100,000 个样本：+8.60）。这种在不同规模下的稳定性突显了 ASFT 的可扩展性，并解决了限制 DFT 在这些场景下的分布漂移问题。对于数学推理任务，DFT 和 ASFT 都超越了标准 SFT，但 ASFT 始终保持一致的优势和更高的训练稳定性。在 100,000 个样本时，ASFT 比基线模型提升+17.89 分（相比之下 DFT 为+13.43），在挑战性基准如 AMC23 上，优势更加明显（36.72% vs. DFT 的 27.19%），反映了其优越的泛化能力。总体而言，ASFT 的改进不仅更大，而且在各种基准测试中更加一致，而 DFT 的提升则更加不稳定——尤其是在需要广泛数学推理的任务中，这突显了 ASFT 的鲁棒性和有效性。

### 5.3 Ablation Study

### 5.3 消融实验

We conduct two ablations: (1) comparing forward vs. reverse KL regularization, and (2) analyzing the impact of learning rate and batch size. These studies clarify the effect of KL direction and the robustness of ASFT to key hyperparameters.

我们进行了两个消融实验：(1) 比较前向与逆向 KL 正则化，以及(2) 分析学习率和批量大小的影响。这些研究澄清了 KL 方向的影响以及 ASFT 对关键超参数的鲁棒性。

**Forward vs. Reverse KL.** Here we follow the standard notation: let $Q$ denote the policy model (i.e., the fine-tuned model with parameters $\theta$), and $P$ denote the reference model (the pretrained base model). In our approach, we use the **forward** KL divergence, $D_{KL}(P \mathbin{/\!/} Q)$, to regularize the policy model. For comparison, we also experiment with the **reverse** KL divergence, $D_{KL}(Q \mathbin{/\!/} P)$. As shown in Figure 2, forward KL leads to consistent improvements. Theoretically, forward KL en-courages **mode-covering** behavior, preventing the model from collapsing and ensuring it maintains the broad distribution of the base model. In contrast, reverse KL is **mode-seeking**, which can cause

前向 KL 与反向 KL。在这里我们遵循标准的表示方法：设 Q 表示策略模型（即参数为 θ 的微调模型），P 表示参考模型（预训练的基础模型）。在我们的方法中，我们使用前向 KL 散度 DKL(P ∥Q)来对策略模型进行正则化。为了进行比较，我们还尝试了反向 KL 散度 DKL(Q ∥P)。如图 2 所示，前向 KL 带来了持续的改进。理论上，前向 KL 鼓励覆盖模式的行为，防止模型崩溃，并确保其保持基础模型的广泛分布。相反，反向 KL 则倾向于寻找模式，这可能会导致

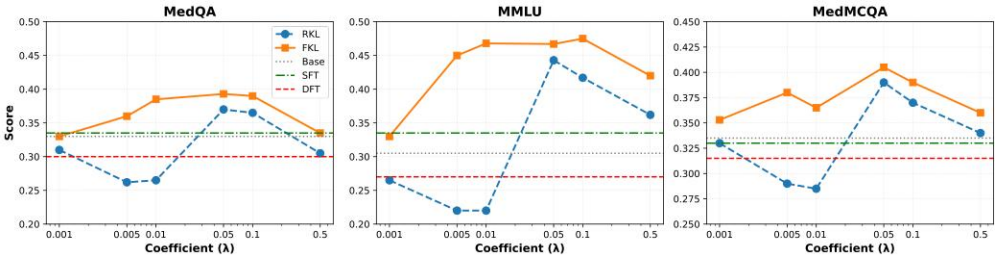


Figure 2: Comparison of forward KL (FKL) and reverse KL (RKL) regularization effects across different coefficient values ($\lambda$) on MedQA, MMLU, and MedMCQA benchmarks. Performance is measured in accuracy, with horizontal dashed lines indicating baseline performance of Base, SFT, and DFT models.

图 2：在 MedQA、MMLU 和 MedMCQA 基准测试中，不同系数值（λ）下前向 KL（FKL）和反向 KL（RKL）正则化效果的比较。性能以准确率衡量，水平虚线表示 Base、SFT 和 DFT 模型的基线性能。

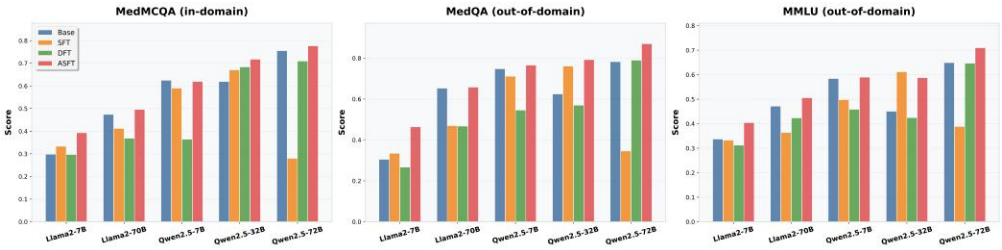


Figure 3: Comparison of model performance across three benchmarks (MedQA, MMLU, MedMCQA) for five models (LLaMA-2-7B, LLaMA-2-70B, Qwen2.5-7B, Qwen2.5-32B, Qwen2.5-72B) using four fine-tuning strategies (Base, SFT, DFT, ASFT). Each subplot shows the scores for a specific benchmark, highlighting the relative effectiveness of different fine-tuning methods across models.

图 3：使用四种微调策略（Base、SFT、DFT、ASFT）对五个模型（LLaMA-2-7B、LLaMA-2-70B、Qwen2.5-7B、Qwen2.5-32B、Qwen2.5-72B）在三个基准测试（MedQA、MMLU、MedMCQA）上进行模型性能比较。每个子图展示了特定基准测试的得分，突出了不同微调方法在不同模型上的相对有效性。

the model to focus excessively on a few high-probability sequences. We further sweep the regularization coefficient $\lambda$ and find that the optimal range is around $\lambda = 0.1$; both excessively small and large values lead to under-anchoring or over-regularization, respectively.

该模型容易过度关注少数高概率序列。我们进一步扫过正则化系数 λ，并发现最优范围大约在 λ=0.1 附近；过小或过大的值分别会导致欠锚定或过度正则化。

**Training Hyper-Parameters Ablation.** We conduct an ablation study to evaluate the robustness of ASFT with respect to learning rate and batch size, using LLaMA-2-7B and 10k samples from

MedCAQA. For learning rate, we examine six values ranging from 5e-6 to 2e-4. ASFT consistently outperforms both SFT and DFT across all configurations, with the best performance observed at intermediate rates (1e-5 and 2e-4). Extreme low (5e-6) or high (2e-4) rates lead to minor drops, highlighting that moderate learning rates are preferable but ASFT remains robust overall. For batch size, we sweep values from 32 to 256. The results show stable performance across the full range, with only small fluctuations. This indicates that batch size is not a sensitive factor for ASFT, and standard settings suffice. Overall, ASFT demonstrates both strong robustness and low sensitivity to key hyperparameters, maintaining a consistent advantage over SFT and DFT in medical QA fine-tuning tasks. The detailed results for learning rate and batch size sweeps are shown in Appendix E.2.

训练超参数消融实验。我们进行了一项消融研究，以评估 ASFT 相对于学习率和批量大小的鲁棒性，使用 LLaMA-2-7B 和 MedCAQA 中的 10,000 个样本。对于学习率，我们检查了从 5e-6 到 2e-4 的六个值。ASFT 在所有配置下都优于 SFT 和 DFT，最佳性能出现在中间范围（1e-5 和 2e-4）。极低（5e-6）或极高（2e-4）的学习率会导致性能略有下降，这表明适度的学习率更优，但 ASFT 总体上仍然具有鲁棒性。对于批量大小，我们尝试了从 32 到 256 的值。结果表明，在整个范围内性能稳定，只有微小波动。这表明批量大小对 ASFT 不是一个敏感因素，标准设置就足够了。总体而言，ASFT 在关键超参数上表现出强大的鲁棒性和低敏感性，在医学问答微调任务中始终保持相对于 SFT 和 DFT 的一致优势。学习率和批量大小的详细结果如附录 E.2 所示。

## 6 Analysis and Discussion

## 6 分析与讨论

### 6.1 Scaling Analysis Across Model Size

### 6.1 跨模型规模的扩展分析

We evaluate fine-tuning methods on medical-domain datasets using 10k training samples (Figure see 3, detailed results in Appendix E.1). Our experiments cover both LLaMA-2 (7B and 70B) and Qwen2.5 (7B, 32B, and 72B) models. Across all model sizes, ASFT consistently outperforms Base, SFT, and DFT, and its improvements remain robust as models scale, demonstrating effective and stable adaptation under low-resource settings.

我们在包含 10,000 个训练样本的医学领域数据集上评估微调方法（见图 3，详细结果见附录 E.1）。我们的实验涵盖了 LLaMA-2（7B 和 70B）和 Qwen2.5（7B、32B 和 72B）模型。在所有模型规模中，ASFT 始终优于 Base、SFT 和 DFT，其改进效果在模型规模扩大时依然稳健，表明其在资源有限的情况下能够有效且稳定地进行适应。

| **Methods** | **MedQA** | **MMLU** | **MedMCQA** | **Avg.** |
|---|---|---|---|---|
| 方法 | MedQA | MMLU | MedMCQA | 平均值 |
| LLaMA-2-7B | 29.85 | 30.52 | 33.76 | 31.38 |
| ***Supervised Fine-Tuning*** | | | | |
| 监督微调 | | | | |
| DFT | 24.67 ↓5.18 | 22.82 ↓7.70 | 30.43 ↓3.33 | 25.97 ↓5.41 |
| iw-SFT | 28.36 ↑1.49 | 35.89 ↑5.37 | 34.88 ↑1.12 | 33.04 ↑1.66 |
| SFT | 33.31 ↑3.56 | 33.52 ↑3.00 | 33.28 ↓0.48 | 33.44 ↑2.06 |
| **ASFT** | **39.28** ↑9.43 | **46.37** ↑15.85 | **40.45** ↑6.69 | **42.03** ↑10.65 |
| ***Reinforcement Learning*** | | | | |
| GRPO | 30.48 ↑0.63 | 32.46 ↑1.94 | 34.64 ↑0.88 | 32.53 ↑1.15 |
| DAPO | 39.75 ↑9.90 | 48.63 ↑18.11 | 38.37 ↑4.61 | 42.25 ↑10.87 |
| 强化学习 | | | | |
| GRPO | 30.48 ↑ 0.63 | 32.46 ↑ 1.94 | 34.64 ↑ 0.88 | 32.53 ↑ 1.15 |
| DAPO | 39.75 ↑ 9.90 | 48.63 ↑ 18.11 | 38.37 ↑ 4.61 | 42.25 ↑ 10.87 |
| ***Continual RL*** | | | | |
| SFT + DAPO | 36.84 ↑6.99 | 44.76 ↑14.24 | 39.11 ↑5.35 | 40.24 ↑8.86 |
| **ASFT + DAPO** | **41.32** ↑11.47 | **49.54** ↑19.02 | **41.45** ↑7.69 | **44.10** ↑12.72 |
| 持续强化学习 | | | | |
| SFT + DAPO | 36.84 ↑ 6.99 | 44.76 ↑ 14.24 | 39.11 ↑ 5.35 | 40.24 ↑ 8.86 |
| ASFT + DAPO | 41.32 ↑ 11.47 | 49.54 ↑ 19.02 | 41.45 ↑ 7.69 | 44.10 ↑ 12.72 |

Table 2: Comparison of different post-training strategies on medical benchmarks. **Bold** numbers indicate the best performance in each group, and rows with blue background highlight our ASFT approach.

表 2：不同后训练策略在医疗基准测试中的比较。粗体数字表示每组中的最佳性能，背景为蓝色的行突出显示了我们的 ASFT 方法。

### 6.2 Comparison with Reinforcement Learning Methods

### 6.2 与强化学习方法的比较

Table 2 presents the performance comparison of different fine-tuning methods on several medical reasoning benchmarks. Our proposed method, **ASFT**, consistently outperforms all SFT-based approaches, including standard SFT, SFT with KL regularization, DFT, and iw-SFT, demonstrating the effectiveness of our adaptive weighting strategy. For example, ASFT achieves an average score of 42.03, which is substantially higher than iw-SFT (33.04) and SFT (33.44). However, as expected, ASFT still falls short of advanced reinforcement learning-based methods such as DAPO, which achieves an average of 42.25, indicating that while our method narrows the gap with RL approaches, RL still maintains a slight advantage in these tasks. The experimental results validate our theoretical analysis from Section 4. The comparison between SFT and DFT (33.44 vs 25.97) confirms DFT's distributional drift problem in knowledge-intensive tasks, while SFT vs ASFT (33.44 vs 42.03) demonstrates the effectiveness of our anchored auxiliary distribution construction. Using final accuracy as a proxy for bound tightness, the performance ordering SFT < ASFT < DAPO (33.44 < 42.03 < 42.25) empirically supports our reward-weighted regression framework, showing that

ASFT achieves a tighter lower bound on the RL objective as proven in Theorem 1. The substantial improvement of ASFT over standard SFT while remaining computationally efficient demonstrates the practical value of our theoretically grounded approach.

表 2 展示了不同微调方法在多个医学推理基准上的性能比较。我们提出的 ASFT 方法在所有基于 SFT 的方法（包括标准 SFT、带 KL 正则化的 SFT、DFT 和 iw-SFT）上表现一致更好，证明了我们自适应加权策略的有效性。例如，ASFT 的平均得分为 42.03，显著高于 iw-SFT（33.04）和 SFT（33.44）。然而，如预期那样，ASFT 仍略逊于基于先进强化学习的方法，如 DAPO，其平均得分为 42.25，这表明尽管我们的方法缩小了与强化学习方法之间的差距，但在这些任务中强化学习仍保持微小优势。实验结果验证了我们第 4 节中的理论分析。SFT 与 DFT（33.44 vs 25.97）的比较确认了 DFT 在知识密集型任务中存在分布漂移问题，而 SFT 与 ASFT（33.44 vs 42.03）的比较则展示了我们锚定辅助分布构建的有效性。使用最终准确率作为边界紧致性的代理指标，性能排序 SFT < ASFT < DAPO（33.44 < 42.03 < 42.25）在实证上支持了我们的奖励加权回归框架，表明 ASFT 在强化学习目标上实现了更紧的下界，正如定理 1 中所证明的那样。ASFT 在保持计算效率的同时显著优于标准 SFT，证明了我们基于理论的方法的实用价值。

### 6.3 ASFT as Enhanced Initialization for Reinforcement Learning

6.3 ASFT 作为强化学习的增强初始化方法

The results in Table 2 further demonstrate that ASFT provides a superior initialization point for sub-sequent RL fine-tuning.Starting from ASFT and continuing training with DAPO yields consistent gains over SFT + DAPO (44.10 vs. 40.24 average, +3.86 points), with the largest improvements on MMLU-medical (+4.78) and MedQA (+4.48). ASFT + DAPO also surpasses standalone DAPO (44.10 vs. 42.25), indicating that KL-anchored fine-tuning not only improves direct supervised per-formance but also creates a more stable policy foundation for RL optimization. This finding sug-gests that the distributional stability provided by ASFT's KL anchoring not only improves direct fine-tuning performance but also creates a more robust foundation for advanced RL algorithms, extending the practical utility of our method beyond standalone applications.

表 2 中的结果进一步表明，ASFT 为后续的强化学习微调提供了更优的初始化点。从 ASFT 开始，并继续使用 DAPO 进行训练，相比 SFT + DAPO（平均 40.24，+3.86 分）能够获得一致的提升，其中在 MMLU-medical 任务上提升最大（+4.78），在 MedQA 任务上也有显著提升（+4.48）。ASFT + DAPO 也优于独立的 DAPO（44.10 vs. 42.25），这表明以 KL 为锚点的微调不仅提升了直接监督下的性能，还为强化学习优化提供了更稳定的策略基础。这一发现表明，ASFT 的 KL 锚定所提供的分布稳定性不仅提升了直接微调的性能，还为高级强化学习算法提供了更稳固的基础，使我们的方法在独立应用之外具有更广泛的实际应用价值。

### 6.4 Cross-Domain Validation on Code Generation

6.4 代码生成中的跨域验证

To examine the generality of ASFT, we fine-tune LLaMA-2-7B on 10k samples from the Magicoder-Evol-Instruct-110K (Wei et al., 2023) dataset using the same setup as Section 5, but for 2 epochs. Evaluation is performed with the evalplus framework (Liu et al., 2023) on HumanEval, Hu-manEval+ (Chen et al., 2021), MBPP, and MBPP+ (Austin et al., 2021). The results are sum-marized in Table 3, ASFT achieves the highest average score, notably improving HumanEval and

为了检验 ASFT 的通用性，我们在与第 5 节相同的设置下，使用 Magicoder-Evol-Instruct-110K（Wei 等，2023）数据集中的 10,000 个样本对 LLaMA-2-7B 进行微调，训练轮数为 2 轮。评估使用 evalplus 框架（Liu 等，2023）在 HumanEval、HumanEval+（Chen 等，

2021）、MBPP 和 MBPP+（Austin 等，2021）上进行。结果总结在表 3 中，ASFT 取得了最高的平均得分，显著提升了 HumanEval 的性能。

| Methods | HumanEval | HumanEval+ | MBPP | MBPP+ | Avg. |
|---|---|---|---|---|---|
| LLaMA-2-7B | 17.1 | 14.6 | 28.3 | 21.2 | 20.3 |
| SFT | 23.2 ↑6.1 | 19.5 ↑4.9 | **34.1** ↑5.8 | **28.6** ↑7.4 | 26.4 ↑6.1 |
| iw-SFT | 23.8 ↑6.7 | 21.3 ↑6.7 | 31.0 ↑2.7 | 26.5 ↑5.3 | 25.7 ↑5.4 |
| DFT | 15.9 ↓1.2 | 12.8 ↓1.8 | 28.3 ±0.0 | 22.3 ↑1.1 | 19.8 ↓0.5 |
| **ASFT** | **27.2** ↑10.1 | **21.4** ↑6.8 | 32.5 ↑4.2 | 26.7 ↑5.5 | **27.0** ↑6.7 |

Table 3: Performance (%) on code generation with LLaMA-2-7B. **Bold** numbers indicate the best performance in each column, and rows with blue background highlight our ASFT approach.

表 3：在使用 LLaMA-2-7B 进行代码生成时的性能（%）。粗体数字表示每列中的最佳性能，背景为蓝色的行突出显示了我们的 ASFT 方法。

HumanEval+ while remaining competitive on MBPP, confirming that the anchoring mechanism gen-eralizes effectively to code generation.

HumanEval+在保持对 MBPP 的竞争力的同时，证实了锚定机制能够有效推广到代码生成中。

### 6.5 Computational Efficiency and Memory Analysis

### 6.5 计算效率与内存分析

While ASFT demonstrates superior performance across both knowledge-intensive and reasoning-intensive domains, the KL divergence computation against the reference model introduces significant computational overhead that merits careful analysis. During training, ASFT requires maintaining the reference model $\pi_{\text{base}}$ in memory alongside the training model $\pi_\theta$, effectively doubling GPU memory consumption from 38.96GB to 88.02GB for full-parameter fine-tuning on LLaMA-2-7B.

尽管 ASFT 在知识密集型和推理密集型领域中表现出色，但其与参考模型之间的 KL 散度计算引入了显著的计算开销，值得仔细分析。在训练过程中，ASFT 需要在内存中同时维护参考模型 $\pi_{\text{base}}$ 和训练模型 $\pi_\theta$，这使得在 LLaMA-2-7B 上进行全参数微调时，GPU 内存消耗从 38.96GB 增加到 88.02GB，几乎翻倍。

The KL computation additionally introduces approximately 23.7% training time overhead compared to standard SFT (0.648 vs 0.524 hours), though remaining substantially more efficient than full RL approaches like GRPO (51.24 hours). All training times are measured on a single NVIDIA A100 GPU. More critically for deployment, inference requires loading both models simultaneously, creating scalability concerns where memory efficiency is paramount. To address these practical limitations, we propose ASFT-LoRA, leveraging the mathematical properties of low-rank adaptation to enable memory-efficient implementation. The key insight exploits LoRA's parameter decomposition $\Delta W = BA$ where the fine-tuned model becomes $\pi_\theta(y|x) = \pi_{\text{base}}(y|x; W_{\text{base}} + BA)$. This parameterization enables computing $D_{\text{KL}}(\pi_{\text{base}}(\cdot|s) \,/\!/\, \pi_\theta(\cdot|s))$ using a single model instance by dynamically switching between $W_{\text{base}}$ and $W_{\text{base}} + BA$ during for-ward passes, eliminating the need for separate model copies while preserving theoretical anchoring

| Method | Time (hrs) | Memory (GB) | Acc |
|---|---|---|---|
| SFT | 0.524 | 38.96 | 33.04 |
| DFT | **0.521** | **38.90** | 25.97 |
| iw-SFT | 8.287 | 100.05 | 33.04 |
| GRPO | 51.24 | 483.98 | 32.53 |
| DAPO | 21.595 | 488.26 | **42.25** |
| ASFT-LoRA | **0.594** | **40.70** | 39.45 |
| ASFT | 0.648 | 88.02 | **42.03** |

Table 4: Training time, memory, and accuracy of different methods on LLaMA-2-7B.

guarantees. As shown in Table 4, ASFT-LoRA (rank=8, lr=5e-4) reduces memory consumption to 40.70GB (comparable to standard SFT) and training time to 0.594 hours (13.4% overhead over SFT), while achieving substantial improvements over SFT baselines (39.45 vs 33.04 overall accuracy). Notably, ASFT-LoRA with appropriate learning rate selection achieves 93.9% of full-parameter ASFT's performance (39.45 vs 42.03 overall accuracy) while requiring less than half the mem-ory, demonstrating the versatility of our anchoring framework across different parameter-efficiency regimes. We observe that LoRA-based training benefits from higher learning rates (5e-4) compared to full fine-tuning (2e-5), consistent with recent findings on low-rank adaptation (Hu et al., 2021).

参数化使得通过在前向传播过程中动态地在 Wbase 和 Wbase + BA 之间切换，使用单个模型实例即可计算 DKL(π base(・|s)∥π θ(・|s))，从而消除了对单独模型副本的需求，同时保持理论锚定保证。如表 4 所示，ASFT-LoRA（rank=8, lr=5e-4）将内存消耗降低至 40.70GB（与标准 SFT 相当），并将训练时间缩短至 0.594 小时（比 SFT 高出 13.4%的开销），同时在整体准确率方面显著优于 SFT 基线（39.45 vs 33.04）。值得注意的是，通过适当选择学习率，ASFT-LoRA 的性能达到了完整参数 ASFT 的 93.9%（39.45 vs 42.03 整体准确率），而所需的内存不到一半，这展示了我们的锚定框架在不同参数效率范围内的灵活性。我们观察到，基于 LoRA 的训练相比完整微调（2e-5）在更高的学习率（5e-4）下表现出更好的效果，这与近期关于低秩适应的研究结果一致（Hu 等，2021）。

### 6.6 Alternative Anchoring Mechanisms.

### 6.6 替代锚定机制

Our theoretical framework reveals that the key to stable optimization lies in providing distributional anchoring to prevent drift, and in principle, any regularization term that constrains the policy dis-tribution can serve this purpose. While KL divergence provides effective anchoring, we investigate whether other terms—particularly the standard SFT loss itself—can offer a more lightweight alter-native. The SFT loss $L_{\mathrm{SFT}}(\theta) = -\mathrm{E}_{(x,y_*)\sim D}[\log \pi_\theta(y_*|\, x)]$ naturally encourages the model to stay close to demonstrated behaviors, suggesting it may provide implicit anchoring effects as observed in Figure 1.

我们的理论框架揭示了稳定优化的关键在于提供分布锚定以防止漂移，原则上，任何限制策略分布的正则化项都可以实现这一目的。虽然 KL 散度提供了有效的锚定，但我们研究了其他项——特别是标准的 SFT 损失本身——是否可以提供一种更轻量的替代方案。SFT 损失 LSFT(θ) = −E(x,y*)~D[log π θ(y*| x)]自然鼓励模型保持在展示行为附近，这表明它可能像图 1 中观察到的那样提供隐式的锚定效果。

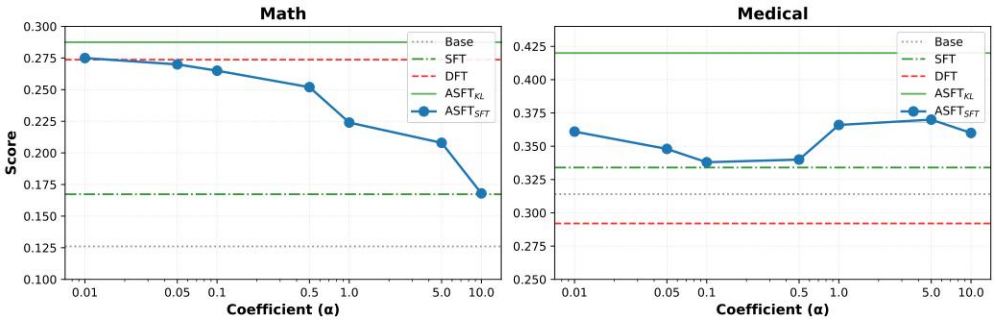


Figure 4: Comparison of KL-based anchoring (ASFT$_{KL}$) versus SFT loss-based anchoring (ASFT$_{SFT}$) across different coefficient values ($\alpha$ or $\lambda$) on math reasoning and medical knowledge tasks. ASFT$_{KL}$ maintains stable performance across all values, while ASFT$_{SFT}$ shows domain-specific sensitivity: degradation at large coefficients for math tasks and volatility for medical tasks.

图 4：在数学推理和医学知识任务上，基于 KL 的锚定（ASFTKL）与基于 SFT 损失的锚定（ASFTSF T）在不同系数值（α或λ）下的比较。ASFTKL 在所有值下保持稳定性能，而 ASFTSF T 表现出领域特异性敏感性：在数学任务中大系数时性能下降，在医学任务中则表现出波动性。

We therefore explore a hybrid objective that combines DFT's adaptive reweighting with SFT-based anchoring:

$$L_{\text{ASFT-SFT}}(\theta) = L_{\text{DFT}}(\theta) + \alpha L_{\text{SFT}}(\theta) \tag{7}$$

因此，我们探索一种混合目标函数，结合了 DFT 的自适应重加权和基于 SFT 的锚定机制：

$$\text{LASFT-SFT}(\theta) = \text{LDFT}(\theta) + \alpha\,\text{LSFT}(\theta) \tag{7}$$

where $\alpha$ controls the anchoring strength. Compared to KL-based anchoring, this approach is compu-tationally more efficient as it does not require maintaining a separate reference model during train-ing. As shown in Figure 4, we conduct a grid search over $\alpha \in \{0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0\}$ on both math and medical tasks.

其中α控制锚定强度。与基于 KL 的锚定方法相比，这种方法在计算上更高效，因为它在训练过程中不需要维护单独的参考模型。如图 4 所示，我们在数学和医学任务上对α ∈ {0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0} 进行网格搜索。

Interestingly, we observe domain-specific patterns: on mathematical reasoning tasks, ASFT-SFT achieves competitive performance at moderate $\alpha$ values (0.05-0.5), but degrades significantly at larger coefficients (dropping from 0.275 to 0.168 as $\alpha$ increases from 0.01 to 10.0). This suggests that for reasoning tasks where the base model already possesses strong prior knowledge, moder-ate SFT anchoring can effectively stabilize training by activating these capabilities without over-constraining exploration. In contrast, on medical knowledge tasks, ASFT-SFT shows more volatile behavior across different $\alpha$ values, with performance fluctuating between 0.340 and 0.370. This instability likely reflects the challenge of balancing knowledge injection (requiring departure from the base distribution) with stability preservation in knowledge-intensive domains. Throughout both settings, ASFT with KL anchoring (green lines) maintains consistently superior and stable perfor-mance across all coefficient values, demonstrating that while SFT loss provides useful anchoring, KL divergence offers more robust distributional control, particularly for tasks requiring significant distribution shift from the base model.

有趣的是，我们观察到一些领域特定的模式：在数学推理任务中，ASFT-SFT 在中等 α 值（0.05-0.5）下表现出竞争力的性能，但在更大的系数下显著退化（当 α 从 0.01 增加到 10.0 时，性能从 0.275 下降到 0.168）。这表明，在基础模型已经具备强大先验知识的推理任务中，适度的 SFT 锚定可以有效稳定训练，通过激活这些能力而不会过度限制探索。相反，在医学知识任务中，ASFT-SFT 在不同 α 值下的表现更加波动，性能在 0.340 和 0.370 之间波动。这种不稳定性可能反映了在知识密集型领域中，平衡知识注入（需要偏离基础分布）与稳定性保持的挑战。在整个设置中，带有 KL 锚定的 ASFT（绿色线条）在所有系数值下都保持了一致的优越和稳定的性能，这表明虽然 SFT 损失提供了有用的锚定，但 KL 散度提供了更稳健的分布控制，特别是在需要从基础模型进行显著分布迁移的任务中。

## 7 CONCLUSION

## 7 结论

We present Anchored Supervised Fine-Tuning (ASFT), a principled approach that addresses the fundamental trade-off between supervised fine-tuning's efficiency and reinforcement learning's gen-eralization. By grounding Dynamic Fine-Tuning in the reward-weighted regression framework, we show that DFT achieves tighter RL lower bounds than SFT but suffers from distributional drift. ASFT resolves this through lightweight KL anchoring, preserving tightness while ensuring stability. Empirically, ASFT consistently outperforms SFT and DFT across mathematical reasoning, medical knowledge injection, and code generation. The method achieves substantial improvements with minimal computational overhead, demonstrating that principled theoretical analysis can lead to both stronger guarantees and practical gains.

我们提出了锚定监督微调（ASFT），这是一种解决监督微调效率与强化学习泛化能力之间根本权衡的原理性方法。通过将动态微调（DFT）建立在奖励加权回归框架中，我们表明 DFT 能够实现比监督微调（SFT）更紧的强化学习下界，但会受到分布漂移的影响。ASFT 通过轻量级 KL 锚定来解决这一问题，在保持紧致性的同时确保稳定性。实验证明，ASFT 在数学推理、医疗知识注入和代码生成等方面均优于 SFT 和 DFT。该方法在计算开销极小的情况下取得了显著的性能提升，表明基于原理性的理论分析可以带来更强的保证和实际效益。

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## ETHICS STATEMENT
## 伦理声明

All experiments in this work were conducted using publicly available datasets and standard benchmarks. The proposed Anchored Supervised Fine-Tuning (ASFT) method is intended for research purposes to improve model generalization and stability. When deploying models trained with ASFT in high-stakes applications, we recommend thorough validation, human oversight, and appropriate safeguards to ensure outputs are accurate, reliable, and aligned with ethical guidelines. We acknowl-edge potential risks of harmful or biased content and encourage the use of input filtering, output moderation, and ongoing monitoring to mitigate such risks. Our goal is to contribute to robust, efficient, and trustworthy AI systems that can be safely deployed.

本工作中所有实验均使用公开数据集和标准基准进行。所提出的锚定监督微调（ASFT）方法旨在改进模型的泛化能力和稳定性，用于研究目的。在将使用 ASFT 训练的模型部署到高风险应用中时，我们建议进行彻底的验证、人工监督以及适当的安全措施，以确保输出的准确性、可靠性和符合伦理准则。我们承认有害或有偏见内容的潜在风险，并鼓励使用输入过滤、输出审核和持续监控来减轻这些风险。我们的目标是为安全部署做出贡献，构建稳健、高效且可信的 AI 系统。

## REPRODUCIBILITY STATEMENT
## 可重复性声明

We provide comprehensive implementation details in Section 5, including specific hyperparameters, datasets, and evaluation protocols. All experiments use publicly available datasets and standard benchmarks. We have released our implementation code to ensure reproducibility of our results. The code is available athttps://github.com/zhuchichi56/ASFT.

我们在第 5 节提供了全面的实现细节，包括具体的超参数、数据集和评估协议。所有实验均使用公开可用的数据集和标准基准。我们已发布实现代码以确保结果的可重复性。代码可在以下链接获取：https://github.com/zhuchichi56/ASFT.

## LIMITATIONS

局限性

We evaluate ASFT on three domains—mathematical reasoning, medical knowledge, and code generation. Although results are consistent across these areas, broader evaluation across diverse task types and domains is needed to fully establish the method's generalizability.

我们在三个领域评估了 ASFT——数学推理、医学知识和代码生成。尽管这些领域的结果具有一致性，但为了充分确立该方法的泛化能力，还需要在多样化的任务类型和领域上进行更广泛的评估。

Our theoretical analysis shows that DFT's effectiveness is fundamentally dependent on the prior knowledge embedded in the base model, which accounts for its strong performance in reasoning tasks with well-established priors, but weaker results in knowledge-intensive domains requiring more substantial distributional shifts. Although our KL-based anchoring mitigates DFT's instability, the method's reliance on model priors persists. Our ablation study on alternative anchoring mechanisms (Section 6.6) suggests that domain-adaptive anchoring strategies could enable more robust generalization across domains; however, developing a systematic framework for such adaptation remains an open challenge.

我们的理论分析表明，DFT 的有效性从根本上依赖于基模型中嵌入的先验知识，这解释了其在具有明确先验知识的推理任务中表现出色，但在需要更大分布偏移的知识密集型领域中表现较弱。尽管我们基于 KL 的锚定方法在一定程度上缓解了 DFT 的不稳定性，但该方法对模型先验知识的依赖仍然存在。我们在替代锚定机制上的消融实验（第 6.6 节）表明，领域自适应的锚定策略可以实现更稳健的跨领域泛化；然而，开发一个系统的适应框架仍然是一个开放性的挑战。

Additionally, ASFT incurs higher computational costs than standard SFT due to the KL divergence computation, requiring doubled GPU memory (from 38.96GB to 88.02GB for LLaMA-2-7B) and 23.7% training time overhead. Our ASFT-LoRA variant (Section 6.5) provides a practical solution by reducing memory to 40.70GB with only 7.3% overhead, though with some performance trade-off.

此外，由于需要计算 KL 散度，ASFT 的计算成本高于标准的 SFT，导致 GPU 内存需求翻倍（从 LLaMA-2-7B 的 38.96GB 增加到 88.02GB），并带来 23.7%的训练时间开销。我们的 ASFT-LoRA 变体（第 6.5 节）通过仅增加 7.3%的开销，将内存需求降低到 40.70GB，提供了一种实用的解决方案，尽管会带来一定的性能折衷。

Our work lacks systematic analysis of what capabilities or behaviors SFT may lose when transitioning to ASFT. While we observe consistent improvements, a more comprehensive understanding of the trade-offs involved in our reweighting scheme would provide valuable insights for practitioners.

我们的工作缺乏对 SFT 在转向 ASFT 过程中可能失去哪些能力和行为的系统分析。虽然我们观察到一致的改进，但对我们重加权方案中涉及的权衡有更全面的理解，将为实践者提供宝贵的见解。

## FUTURE WORK

未来工作

A promising direction for future research is extending ASFT to continual learning settings, where models must adapt to new tasks while preserving previously acquired knowledge. The anchor-ing mechanism in ASFT naturally aligns with continual learning goals: the KL regularization that prevents distributional drift during single-task fine-tuning could be adapted to prevent catastrophic forgetting when learning sequential tasks. By maintaining anchors to both the base model and

未来研究的一个有前景方向是将 ASFT 扩展到持续学习场景中，在这种场景下，模型必须在保持之前学到的知识的同时适应新任务。ASFT 中的锚定机制自然地与持续学习目标相一致：在单任务微调过程中，防止分布漂移的 KL 正则化可以被调整以防止在学习序列任务时出现灾难性遗忘。通过同时维护基础模型和锚定机制，

checkpoints from previous tasks, ASFT's framework could potentially enable stable, progressive capability acquisition across multiple domains.

来自先前任务的检查点，ASFT 的框架有可能在多个领域实现稳定、逐步的能力获取。

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## A LLM USAGE

## 大语言模型的使用

In the preparation of this paper, we only used large language models (LLMs) as an assistive tool for grammar correction and text polishing.

在撰写本文的过程中，我们仅将大型语言模型（LLMs）作为辅助工具，用于语法修正和文本润色。

## B LIMITATIONS

## B 局限性

We evaluate ASFT on three domains—mathematical reasoning, medical knowledge, and code generation. Although results are consistent across these areas, broader evaluation across diverse task types and domains is needed to fully establish the method's generalizability.

我们在三个领域评估了 ASFT——数学推理、医学知识和代码生成。尽管这些领域的结果具有一致性，但为了充分确立该方法的泛化能力，还需要在多样化的任务类型和领域上进行更广泛的评估。

Our work lacks systematic analysis of what capabilities or behaviors SFT may lose when transitioning to ASFT. While we observe consistent improvements, a more comprehensive understanding of the trade-offs involved in our reweighting scheme would provide valuable insights for practitioners.

我们的工作缺乏对 SFT 在转向 ASFT 过程中可能失去哪些能力和行为的系统性分析。虽然我们观察到一致的改进，但对我们的重加权方案所涉及的权衡有更全面的理解，将为实践者提供宝贵的见解。

## C BROADER IMPACTS

## C 更广泛的影响

ASFT offers a lightweight and computationally efficient alternative to standard supervised fine-tuning, requiring only KL regularization rather than complex reward modeling. This simplicity makes advanced fine-tuning techniques more accessible to researchers and practitioners with limited computational resources. The method shows potential as a practical path forward for improving language model post-training, particularly in specialized domains requiring both knowledge injection and reasoning capabilities. However, as with any fine-tuning approach, careful evaluation and validation remain essential when deploying models in high-stakes applications such as medical or educational settings.

ASFT 提供了一种轻量且计算效率高的替代方案，用于标准的监督微调，只需使用 KL 正则化，而无需复杂的奖励建模。这种简单性使得先进的微调技术更易于计算资源有限的研究人员和实践者使用。该方法在提高语言模型微调方面展现出潜力，特别是在需要知识注入和推理能力的专业领域。然而，如同任何微调方法一样，在将模型部署到高风险应用（如医疗或教育环境）中时，仍需仔细评估和验证。

## D THEORETICAL FOUNDATIONS

## D 理论基础

This appendix provides detailed derivations for the key theoretical results presented in Section 3.

本附录提供了第 3 节中呈现的关键理论结果的详细推导。

## D.1 Derivation of SFT as RL Lower Bound

D.1 SFT 作为 RL 下界的推导

We provide a complete proof of Proposition 1, following the theoretical framework established in (Qin & Springenberg, 2025), showing that SFT optimizes a lower bound on the RL objective in sparse reward settings.

我们提供了命题 1 的完整证明，遵循(Qin & Springenberg, 2025)中建立的理论框架，证明了 SFT 在稀疏奖励设置下优化了 RL 目标的下界。

*Proof of Proposition 1.* We start with the RL objective under sparse rewards $R(\tau) = \mathrm{I}[y = y_*]$:

命题 1 的证明。我们从稀疏奖励下的 RL 目标 $R(\tau) = \mathrm{I}[y = y^*]$开始：

$$J(\theta) = \mathrm{E}_{\tau\sim\pi_\theta}[R(\tau)] = \mathrm{E}_{\tau\sim\pi_\theta}[\mathrm{I}[y = y_*]] \tag{8}$$

J( θ ) = E_{ τ ~ π θ }[R( τ )] = E_{ τ ~ π θ }[I[y = y*]] (8)

Since we only observe trajectories from the reference distribution $\pi_{\text{ref}}$, we use importance sampling to rewrite this expectation. Under the assumption $\mathrm{supp}(\pi_\theta) \subseteq \mathrm{supp}(\pi_{\text{ref}})$:

由于我们只能观察来自参考分布 πref 的轨迹，我们使用重要性采样来重写这个期望。在假设 supp( π θ ) ⊆ supp( π ref) 的前提下：

$$J(\theta) = \mathrm{E}_{\tau\sim\pi_{\text{ref}}} \frac{\pi_\theta(\tau)}{\pi_{\text{ref}}(\tau)} \mathrm{I}[y = y_*] \tag{9}$$

J( θ ) = E_{ τ ~ π _ref}

Now we apply the fundamental inequality $u \geq 1 + \log u$ for all $u > 0$. Setting $u = \frac{\pi_\theta(\tau)}{\pi_{\text{ref}}(\tau)}$:

$$\pi_{\text{ref}}(\tau) \geq 1 + \log \pi_\theta(\tau)\pi_{\text{ref}}(\tau) = 1 + \log \pi_\theta(\tau) - \log \pi_{\text{ref}}(\tau) \tag{10}$$

Substituting this into our importance-sampled expression:

将这个代入我们的重要性采样表达式中：

$$J(\theta) \geq \mathrm{E}_{\tau\sim\pi_{\text{ref}}} [(1 + \log \pi_\theta(\tau) - \log \pi_{\text{ref}}(\tau)) \mathrm{I}[y = y_*]] \tag{11}$$

$$= \mathrm{E}_{\tau\sim\pi_{\text{ref}}}[\mathrm{I}[y = y_*]] + \mathrm{E}_{\tau\sim\pi_{\text{ref}}}[\mathrm{I}[y = y_*] \log \pi_\theta(\tau)] \tag{12}$$

$$- \mathrm{E}_{\tau\sim\pi_{\text{ref}}}[\mathrm{I}[y = y_*] \log \pi_{\text{ref}}(\tau)] \tag{13}$$

J( θ ) ≥ E_{ τ ~ π _ref} [(1 + log π _ θ ( τ ) − log π _ref( τ )) I[y = y^*]] (11)
= E_{ τ ~ π _ref}[I[y = y^*]] + E_{ τ ~ π _ref}[I[y = y^*] log π _ θ ( τ )] (12)
− E_{ τ ~ π _ref}[I[y = y^*] log π _ref( τ )] (13)

The first and third terms are constants independent of $\theta$. Let $c_{\text{ref}} = \mathrm{E}_{\tau\sim\pi_{\text{ref}}}[\mathrm{I}[y = y_*]] = \mathrm{P}_{\pi_{\text{ref}}}(\tau \in D_+)$. Then:

$$J(\theta) \geq c_{\text{ref}} + \mathrm{E}_{\tau\sim\pi_{\text{ref}}}[\mathrm{I}[y = y_*] \log \pi_\theta(\tau)] + \text{const} \tag{14}$$

第一个和第三个项是与 θ 无关的常数。令 cref = E τ ~ π ref[I[y = y*]] = P π ref( τ ∈D+)。则：

J( θ ) ≥ cref + E τ ~ π ref[I[y = y*] log π θ ( τ )] + const (14)

The expectation over indicator-weighted trajectories can be rewritten as an expectation over the
指标加权轨迹上的期望可以重写为对的期望

filtered dataset $D_+ = \{(x, y_*) \mid R(x, y_*) = 1\}$:

过滤后的数据集 D+ = {(x, y*) | R(x, y*) = 1}:

$$\mathrm{E}_{\tau \sim \pi_{\mathrm{ref}}}[\mathrm{I}[y = y_*] \log \pi_\theta(\tau)] = c_{\mathrm{ref}} \mathrm{E}_{\tau \in D_+}[\log \pi_\theta(\tau)] \quad (15)$$

E_{ τ ~ π_{ref}}[I[y = y^*] log π_θ(τ)] = c_{ref}E_{ τ ∈D^+}[log π_θ(τ)] (15)

Dropping the constant terms, we obtain:
去掉常数项，我们得到：

$$J(\theta) \geq c_{\mathrm{ref}} \mathrm{E}_{\tau \in D_+}[\log \pi_\theta(\tau)] \quad (16)$$

This completes the proof. Note that the right-hand side is precisely the SFT objective (up to scaling),
这完成了证明。请注意，右边的表达式正是 SFT 目标（在缩放意义下）。

establishing that SFT optimizes a lower bound on the RL objective.
建立 SFT 优化 RL 目标的下界。

### D.2 Frozen Auxiliary Distribution and Optimization Validity

D.2 冻结辅助分布和优化有效性

We formalize how DFT's stop-gradient mechanism ensures valid lower bound optimization by treat-
我们形式化了 DFT 的 stop-gradient 机制如何通过处理来确保有效的下界优化

ing the auxiliary distribution as constant during each update step.
在每次更新步骤中将辅助分布视为常数。

**Lemma 1** (Frozen-q Surrogate)**.** *At iteration k, define the auxiliary distribution*
引理 1（冻结 q 替代函数）。在第 k 次迭代时，定义辅助分布

$$q_k(\tau) \propto \pi_{\mathrm{ref}}(\tau \mid D_+)\, p_{\theta_k}(\tau) \text{ where} \quad (17)$$

*$\theta_k$ is the current parameter. Then optimizing*

$$\max_\theta \mathrm{E}_{\tau \in D_+} \frac{q_k(\tau)}{\pi_{\mathrm{ref}}(\tau)} \log \pi_\theta(\tau) \quad (18)$$

*with $q_k$ treated as constant (independent of $\theta$) yields a valid lower bound on $J(\theta)$ that differs only by a $\theta$-independent constant.*

将 qk 视为常数（与 θ 无关）时，可以得到一个有效的 J(θ)下界，该下界仅与 θ 无关的常数不同。

*Proof.* Following the derivation in Appendix D.3, we have:
证明。根据附录 D.3 中的推导，我们有：

$$
J(\theta) \geq \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \left[ \log \frac{\pi_\theta(\tau)}{q_k(\tau)} \right] - \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \left[ \log \frac{\pi_{\text{ref}}(\tau) R(\tau)}{q_k(\tau)} \right] \quad (19)
$$

When $q_k$ is constructed from frozen parameter $\theta_k$ (via stop-gradient), the first and third terms become constants independent of the optimization variable $\theta$. Therefore, maximizing the second term is equivalent to maximizing a valid lower bound on $J(\theta)$.

当 qk 由冻结参数 θ k（通过 stop-gradient）构造时，第一项和第三项成为与优化变量 θ 无关的常数。因此，最大化第二项等价于最大化 J( θ )的一个有效下界。

The stop-gradient operator sg[·] in DFT implements this frozen-$q$ mechanism: at each iteration, $q_k$ depends on $\theta_k$ from the previous step, but is treated as fixed when computing gradients with respect to the current $\theta$.

DFT 中的停止梯度运算符 sg[・]实现了这种冻结 q 机制：在每次迭代中，qk 依赖于上一步的 θ k，但在计算当前 θ 的梯度时，将其视为固定值。

This formalization clarifies that the ”dropped constant terms” in our derivations are legitimate because they are genuinely $\theta$-independent within each optimization iteration, making DFT a valid iterative lower bound maximization procedure.

这种形式化说明了我们在推导中所说的“被舍去的常数项”是合理的，因为它们在每次优化迭代中确实是真正与 θ 无关的，从而使 DFT 成为一种有效的迭代下界最大化过程。

### D.3 From Importance Sampling to Importance-Weighted Lower Bounds

D.3 从重要性采样到重要性加权下界

We now derive the generalized importance-weighted framework that enables tighter bounds through auxiliary distributions.

我们现在推导出一个广义的重要性加权框架，通过辅助分布实现更紧密的界限。

Starting from the importance-sampled RL objective:

$$
J(\theta) = \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \left[ \frac{\pi_\theta(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) \right] \quad (20)
$$

从重要性采样的强化学习目标出发：

$$
J(\theta) = \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \left[ \frac{\pi_\theta(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) \right] \quad (20)
$$

We introduce an auxiliary distribution $q(\tau)$ with $\text{supp}(q) \supseteq \text{supp}(\pi_\theta)$ and rewrite the importance ratio:

$$
J(\theta) = \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \left[ \frac{\pi_\theta(\tau)}{q(\tau)} \cdot \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) \right] \quad (21)
$$

我们引入一个辅助分布 $ q(\tau) $，使得 $ \text{supp}(q) \supseteq \text{supp}(\pi_\theta) $，并将重要性比率重新表示为：

$$
J(\theta) = \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \left[ \frac{\pi_\theta(\tau)}{q(\tau)} \cdot \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) \right] \quad (21)
$$

Now we apply the inequality $u \geq 1 + \log u$ to the ratio $\frac{\pi_\theta(\tau)}{q(\tau)}$:

$$\frac{\pi_\theta(\tau)}{q(\tau)} \geq 1 + \log \frac{\pi_\theta(\tau)}{q(\tau)} = 1 + \log \pi_\theta(\tau) - \log q(\tau) \tag{22}$$

现在我们把不等式 $u \geq 1 + \log u$ 应用于比值 $\frac{\pi_\theta(\tau)}{q(\tau)}$:

$$\frac{\pi_\theta(\tau)}{q(\tau)} \geq 1 + \log \frac{\pi_\theta(\tau)}{q(\tau)} = 1 + \log \pi_\theta(\tau) - \log q(\tau) \quad (22)$$

Substituting this bound:

$$J(\theta) \geq \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} (1 + \log \pi_\theta(\tau) - \log q(\tau)) R(\tau) \tag{23}$$

$$= \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) + \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) \log \pi_\theta(\tau) \tag{24}$$

$$- \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) \log q(\tau) \tag{25}$$

将这个界代入:

$$J(\theta) \geq \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} (1 + \log \pi_\theta(\tau) - \log q(\tau)) R(\tau) \tag{23}$$

$$= \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) + \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) \log \pi_\theta(\tau) \tag{24}$$

$$- \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) \log q(\tau) \tag{25}$$

When treating $q$ as fixed (independent of $\theta$, see Lemma 1), the first and third terms are constants, so

当将 q 视为固定（与 θ 无关，参见引理 1）时，前两个项是常数，因此

we can drop them from the optimization objective:

$$J(\theta) \geq \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} R(\tau) \log \pi_\theta(\tau) + \text{const} \tag{26}$$

For sparse rewards $R(\tau) = \mathbb{I}[y = y_*]$, this reduces to:

$$J(\theta) \geq \mathbb{E}_{\tau \sim \pi_{\text{ref}}} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} \mathbb{I}[y = y_*] \log \pi_\theta(\tau) + \text{const} \tag{27}$$

Converting to an expectation over the filtered dataset $D_+$:

$$J(\theta) \geq c_{\text{ref}} \mathbb{E}_{\tau \in D_+} \frac{q(\tau)}{\pi_{\text{ref}}(\tau)} \log \pi_\theta(\tau) + \text{const} \tag{28}$$

where $c_{\text{ref}} = \mathbb{P}_{\pi_{\text{ref}}}(\tau \in D_+)$.

**Key Insights:**

其中 $c_{\text{ref}} = \mathbb{P}_{\pi_{\text{ref}}}(\tau \in D_+)$.
关键见解:

1. When $q(\tau) = \pi_{\text{ref}}(\tau)$, we recover the standard SFT bound from Proposition 1.

1. 当 $q(\tau) = \pi_{ref}(\tau)$ 时，我们可以从命题 1 中恢复出标准的 SFT 界。

2. As $q(\tau) \to \pi_\theta(\tau)$, the bound becomes tighter since the inequality $\frac{\pi_\theta(\tau)}{q(\tau)} \geq 1 + \log \frac{\pi_\theta(\tau)}{}$ approaches equality.

2. 当 $q(\tau)$ 趋近于 $\pi_\theta(\tau)$ 时，由于不等式 $\frac{\pi_\theta(\tau)}{q(\tau)} \geq 1 + \log \frac{\pi_\theta(\tau)}{}$ 趋于等式，因此界限变得更紧。

3. The choice of $q$ involves a fundamental trade-off: tighter bounds (by making $q$ closer to $\pi_\theta$) versus stability (by keeping $q$ close to $\pi_{ref}$).

3. 选择 q 涉及一个根本性的权衡：更紧的界限（通过使 q 更接近 $\pi_\theta$）与稳定性（通过保持 q 接近 $\pi_{ref}$）之间的平衡。

This theoretical framework provides the foundation for both existing methods like DFT and our pro-posed ASFT approach, which aims to achieve tight bounds while maintaining optimization stability through principled anchoring mechanisms.

这一理论框架为现有的方法（如 DFT）以及我们提出的 ASFT 方法奠定了基础，该方法旨在通过原理性的锚定机制，在保持优化稳定性的同时实现紧致的界限。

## D.4 Proof of Tightness

D.4 紧性证明

We provide the detailed proof that the DFT auxiliary distribution $q$ yields a strictly tighter lower bound than standard SFT whenever the policy distribution is non-degenerate.

我们提供了详细的证明，当策略分布非退化时，DFT 辅助分布 q 所提供的下界比标准 SFT 更严格。

*Proof of Theorem 1.* Let $X = p_\theta(\tau)$ with $\tau \sim \pi_{ref}(\cdot \mid D_+)$, and $f(x) = \log x$. We compare:

$$B_{SFT} = c_{ref}\,\mathbb{E}[f(X)], \qquad B_{DFT} = c_{ref}\,\frac{\mathbb{E}[Xf(X)]}{\mathbb{E}[X]} \tag{29}$$

定理 1 的证明。设 $X = p_\theta(\tau)$，其中 $\tau \sim \pi_{ref}(\cdot \mid D_+)$，且 $f(x) = \log x$。我们比较：

$$B_{SFT} = c_{ref}\,\mathbb{E}[f(X)], \qquad B_{DFT} = c_{ref}\,\frac{\mathbb{E}[Xf(X)]}{\mathbb{E}[X]} \tag{29}$$

The difference is:

差异在于：

$$B_{DFT} - B_{SFT} = c_{ref}\left(\frac{\mathbb{E}[Xf(X)]}{\mathbb{E}[X]} - \mathbb{E}[f(X)]\right) \tag{30}$$

$$= \frac{c_{ref}}{\mathbb{E}[X]}\left(\mathbb{E}[Xf(X)] - \mathbb{E}[X]\mathbb{E}[f(X)]\right) \tag{31}$$

$$= c_{ref}\,\mathbb{E}[X]\,\mathrm{Cov}(X, f(X)) \tag{32}$$

Since $f(x) = \log x$ is strictly increasing on $(0, 1]$, variables $X$ and $f(X)$ are comonotone, yielding

由于 $f(x) = \log x$ 在 $(0, 1]$ 上是严格递增的，变量 X 和 f(X) 是共单调的，从而得到

$\mathrm{Cov}(X, f(X)) \geq 0$ with equality iff $X$ is constant. Therefore $B_{\mathrm{DFT}} \geq B_{\mathrm{SFT}}$, with strict inequality

$\mathrm{Cov}(X, f(X)) \geqslant 0$，当且仅当 X 为常数时取等号。因此 BDFT ⩾BSFT，且严格不等式成立

when $\mathrm{Var}(X) > 0$.

当 $\mathrm{Var}(X) > 0$ 时。

# E EXPERIMENTS

## E.1 MODEL SCALE RESULTS
模型规模结果

| Model | Methods | MedQA | MMLU | MedMCQA | Avg. |
|---|---|---|---|---|---|
| LLaMA-2-7B | Base | 29.85 | 30.52 | 33.76 | 31.38 |
| | SFT | 33.31 ↑3.46 | 33.52 ↑3.00 | 33.28 ↓0.48 | 33.37 ↑1.99 |
| | SFT w KL | 29.22 ↓0.63 | 30.63 ↑0.11 | 33.01 ↓0.75 | 30.95 ↓0.43 |
| | iw-SFT | 35.35 ↑5.50 | 38.92 ↑8.40 | 34.74 ↑0.98 | 36.34 ↑4.96 |
| | DFT | 29.69 ↓0.16 | 26.69 ↓3.83 | 31.20 ↓2.56 | 29.19 ↓2.19 |
| | GRPO | 30.48 ↑0.63 | 32.46 ↑1.94 | 34.64 ↑0.88 | 32.53 ↑1.15 |
| | DAPO | **39.75** ↑9.90 | **48.63** ↑18.11 | 38.37 ↑4.61 | **42.25** ↑10.87 |
| | **ASFT** | 39.28 ↑9.43 | 46.37 ↑15.85 | **40.45** ↑6.69 | 42.03 ↑10.65 |
| LLaMA-2-70B | Base | 47.37 | 65.32 | 47.21 | 53.30 |
| | DFT | 36.84 ↓10.53 | 46.77 ↓18.55 | 42.39 ↓4.82 | 42.00 ↓11.30 |
| | SFT | 41.24 ↓6.13 | 47.02 ↓18.30 | 36.43 ↓10.78 | 41.56 ↓11.74 |
| | **ASFT** | **49.57** ↑2.20 | **65.86** ↑0.54 | **50.61** ↑3.40 | **55.35** ↑2.05 |
| Qwen2.5-7B | Base | 62.53 | 74.84 | 58.45 | 65.27 |
| | SFT | 59.07 ↓3.46 | 71.19 ↓3.65 | 49.80 ↓8.65 | 60.02 ↓5.25 |
| | SFT w KL | 61.12 ↓1.41 | 69.77 ↓5.07 | 51.52 ↓6.93 | 60.80 ↓4.47 |
| | DFT | 36.45 ↓26.08 | 54.65 ↓20.19 | 45.90 ↓12.55 | 45.67 ↓19.60 |
| | GRPO | **63.00** ↑0.47 | 76.16 ↑1.32 | 58.93 ↑0.48 | 66.03 ↑0.76 |
| | DAPO | 63.94 ↑1.41 | 73.82 ↓1.02 | 58.88 ↑0.43 | 65.55 ↑0.28 |
| | **ASFT** | 61.98 ↓0.55 | **76.60** ↑1.76 | **59.00** ↑0.55 | **65.86** ↑0.59 |
| Qwen2.5-32B | Base | 61.90 | 62.50 | 45.09 | 56.50 |
| | SFT | 67.09 ↑5.19 | 76.12 ↑13.62 | 61.20 ↑16.11 | 68.14 ↑11.64 |
| | DFT | 68.42 ↑6.52 | 57.03 ↓5.47 | 42.51 ↓2.58 | 55.99 ↓0.51 |
| | **ASFT** | **71.80** ↑9.90 | **79.23** ↑16.73 | **58.76** ↑13.67 | **69.93** ↑13.43 |
| Qwen2.5-72B | Base | 75.57 | 78.28 | 64.91 | 72.92 |
| | SFT | 28.04 ↓47.53 | 34.57 ↓43.71 | 38.92 ↓25.99 | 33.84 ↓39.08 |
| | DFT | 71.01 ↓4.56 | 79.04 ↑0.76 | 64.69 ↓0.22 | 71.58 ↓1.34 |
| | **ASFT** | **77.69** ↑2.12 | **87.08** ↑8.80 | **71.00** ↑6.09 | **78.59** ↑5.67 |

Table 5: Performance of various fine-tuning methods on medical domain datasets for both LLaMA-2 and Qwen2.5 series. **Bold** numbers indicate the best performance in each group, and rows with blue background highlight our ASFT approach. Arrows with ↑and ↓indicate performance improvements and degradations relative to the base model. All values are reported in percentages.

表 5：LLaMA-2 和 Qwen2.5 系列在医疗领域数据集上各种微调方法的性能。粗体数字表示每组中的最佳性能，背景为蓝色的行突出显示我们的 ASFT 方法。↑和↓箭头表示相对于基础模型的性能提升和下降。所有数值均以百分比形式报告。

### E.2 TRAINING HYPER-PARAMETERS

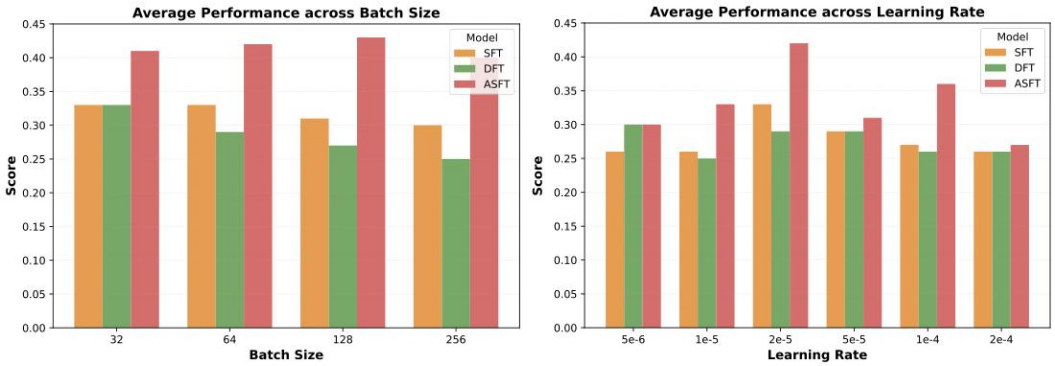


Figure 5: Ablation study of ASFT on MedCAQA (10k) using LLaMA-2-7B. **Right:** Effect of batch size, indicating stable performance across a wide range (32–256). ASFT consistently outperforms SFT and DFT under all settings. **Left:** Effect of different learning rates on average performance, showing that intermediate rates (1e-5 and 1e-4) achieve the best results.

图 5：使用 LLaMA-2-7B 在 MedCAQA（10k）上对 ASFT 进行消融研究。右边：批量大小的影响，表明在较宽范围内（32－256）性能稳定。ASFT 在所有设置下均优于 SFT 和 DFT。左边：不同学习率对平均性能的影响，表明中间的学习率（1e-5 和 1e-4）取得了最佳效果。

# F DATASET EXAMPLE
# F 数据集示例

## F.1 MEDICAL KNOWLEDGE DATASETS
## F.1 医学知识数据集

医学知识训练集

**Instruction:**
Disulfiram is a type of:-
A. Aversion therapy
B. Anticraving therapy
C. Detoxification
D. Opioid management therapy

二硫化物是一种：
A. 戒断疗法
B. 减少欲望疗法
C. 解毒
D. 阿片类药物管理疗法

**Response:**
Deterrent agents/Aversive agents Disulfiram (tetraethyl thiuram disulfide) When alcohol is ingested by a person who is on disulfiram, alcohol-derived acetaldehyde cannot be oxidized to acetate and this leads to an accumulation of acetaldehyde in blood. This causes the impoant disulfiram-ethanol reaction (DER) characterized by flushing, tachycardia, hypotension, tachypnoea, palpitations, headache, sweating, nausea, vomiting, giddiness and a sense of impending doom associated with severe anxiety. When we try to decrease the frequency of behavior because it can cause a really bad consequence It will be called as an aversion or Aversive therapy. Anti-craving agents Acamprosate, naltrexone and SSRIs (such as fluoxetine) are among the medications tried as anti-craving agents in alcohol dependence. Detoxification Detoxification is the treatment of alcohol withdrawal symptoms, i.e. symptoms produced by the removal of the 'toxin' (alcohol). The drugs of choice for detoxification are usually benzodiazepines.

回应：
威慑剂/厌恶剂 二乙基硫代氨基甲酸酯（二乙基硫代氨基甲酸酯） 当服用二乙基硫代氨基甲酸酯的人饮酒时，酒精衍生的乙醛无法被氧化为乙酸，导致血液中乙醛的积累。这会引起重要的二乙基硫代氨基甲酸酯-酒精反应（DER），表现为面部潮红、心率加快、低血压、呼吸急促、心悸、头痛、出汗、恶心、呕吐、头晕以及与严重焦虑相关的即将发生灾难的感觉。当我们试图减少某种行为的频率，因为它会导致非常严重的后果时，这种做法被称为厌恶或厌恶疗法。抗渴求药物 酒精依赖治疗中尝试作为抗渴求药物的药物包括阿坎酸、纳曲酮和 SSRI（如氟西汀）。戒毒 戒毒是指治疗酒精戒断症状，即由去除“毒素”（酒精）所产生的症状。通常用于戒毒的首选药物通常是苯二氮䓬类药物。

So the answer to this question is A

所以这个问题的答案是 A

Figure 6: An example from the medical knowledge training set illustrating an instruction-response pair on alcohol dependence therapy

图 6：来自医学知识训练集的一个示例，展示了酒精依赖治疗中的指令-响应对

医学知识测试集

**Question:**
Which of the following is not true for myelinated nerve fibers:
**Options:**
A. Impulse through myelinated fibers is slower than non-myelinated fibers
B. Membrane currents are generated at nodes of Ranvier
C. Saltatory conduction of impulses is seen
D. Local anesthesia is effective only when the nerve is not covered by myelin sheath
**Answer:** A

问题：
以下哪项不适用于有髓鞘神经纤维：
选项：
A. 有髓鞘纤维的冲动传导速度比无髓鞘纤维慢
B. 网状结构节点产生膜电流
C. 冲动的跳跃式传导可见
D. 局部麻醉只有在神经未被髓鞘覆盖时才有效
答案：A

Figure 7: An example from the medical knowledge test set illustrating a multiple-choice question on physiology (myelinated nerve fibers)

图 7：医学知识测试集中的一个例子，展示了一道关于生理学（有髓神经纤维）的多项选择题

## F.2 MATH REASONING DATASETS

F.2 数学推理数据集

数学推理训练集

**Instruction:**
Let $a$, $b$, and $c$ be the roots of the equation $x^3-2x-5=0$. Find $\frac{1}{a-2}+\frac{1}{b-2}+\frac{1}{c-2}$.

设 a, b 和 c 是方程 $x^3 - 2x - 5 = 0$ 的根，求 $a^{-2} + b^{-2} + c^{-2}$。

**Response:**
First, substitute $x = z + 2$ to transform the polynomial so that $z + 2$ are the new roots, i.e., $a-2$, $b-2$, and $c-2$ are the roots for $z$. We then have:

回答：
首先，将 $x = z + 2$ 代入，将多项式转换为以 $z + 2$ 为新根的形式，即 $a-2$, $b-2$ 和 $c-2$ 是 z 的根。然后我们有：

$$(z+2)^3-2(z+2)-5=z^3+6z^2+12z+8-2z-4-5=z^3+6z^2+10z-1$$

$$(z + 2)^3 - 2(z + 2) - 5 = z^3 + 6z^2 + 12z + 8 - 2z - 4 - 5 = z^3 + 6z^2 + 10z - 1$$

We are interested in $\frac{1}{a-2}+\frac{1}{b-2}+\frac{1}{c-2}$, which are the reciprocals of the roots of $z^3+6z^2+10z-1$. By Vieta's formulas, the sum of the reciprocals of the roots of a polynomial $z^3+pz^2+qz+r$ is $-q/r$. For our polynomial, $p=6$, $q=10$, and $r=-1$, so:

我们关注的是 $a^{-2} + \frac{1}{b^{-2}} + c^{-2}$，它们是多项式 $z^3 + 6z^2 + 10z - 1$ 的根的倒数。根据韦达定理，多项式 $z^3 + pz^2 + qz + r$ 的根的倒数之和为 $-q/r$。对于我们的多项式，$p = 6$，$q = 10$，$r = -1$，因此：

$$\frac{1}{a-2}+\frac{1}{b-2}+\frac{1}{c-2}=-\frac{q}{r}=-\frac{10}{-1}=10$$

$$a^{-2} + b^{-2} + c^{-2} = -q, r = -10^{-1} = 10$$

Figure 8: An example from the math reasoning training set illustrating an instruction-response pair on algebraic root manipulation

图 8：来自数学推理训练集的一个示例，展示了代数根运算的指令-响应对

数学推理测试集

**Question:**
What is the smallest positive perfect cube that can be written as the sum of three consecutive integers?

问题：
能表示为三个连续整数之和的最小正完美立方数是多少？

**Answer:**
27

答案：
27

**Solution:**
The sum of three consecutive integers takes the form $(k-1)+(k)+(k+1)=3k$ and hence is a multiple of 3. Conversely, if a number $n$ is a multiple of 3, then $n/3-1$, $n/3$, and $n/3+1$ are three consecutive integers that sum to give $n$. Therefore, a number is a sum of three consecutive integers if and only if it is a multiple of 3. The smallest positive perfect cube that is a multiple of 3 is $3^3=27$ .

解法：
三个连续整数的和可以表示为 $(k-1)+k+(k+1)=3k$，因此其一定是 3 的倍数。反过来，如果一个数 n 是 3 的倍数，那么 $n/3-1$，$n/3$，$n/3+1$ 这三个连续整数相加的结果就是 n。因此，一个数可以表示为三个连续整数的和，当且仅当它是 3 的倍数。最小的正完美立方数且是 3 的倍数的是 $3^3=27$。

Figure 9: An example from the math reasoning test set illustrating a problem with its solution on sums of consecutive integers

图 9：数学推理测试集中的一个例子，展示了一个关于连续整数和的问题及其解法